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POLAR WANDERING AND CONTINENTAL DRIFT: EVIDENCE FROM PALEOMAGNETIC OBSERVATIONS IN THE UNITED STATES

D. W COLLINSON and S. K RUNCORN

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POLAR WANDERING AND CONTINENTAL DRIFT: EVIDENCE FROM PALEOMAGNETIC OBSERVATIONS IN THE UNITED STATES

BY D. W. COLLINSON AND S. K. RUNCORN

ABSTRACT

Further studies of the paleomagnetic directions of red sandstones and siltstones of various geological ages in the United States are described. Usually the directions of magnetization of samples from a formation at one site are grouped symmetrically about a mean direction. From such a mean direction the position of the pole for that geological age can be calculated. There are, however, magnetically unstable formations in which the directions of magnetization are distributed approximately in the plane containing the present dipole field at the site and the original direction of the magnetic field. This planar distribution is the result of a superposition of a secondary magnetization on the original one. The former is thought to be a viscous or chemical magnetization acquired in the last 1,000–1,000,000 years.

Pole positions calculated from mean directions at different sites are consistent for the same formation and for different formations of the same geological age.

The study confirms the general trend of the polar-wandering curve for North America obtained by Runcorn (1956a), which lies around the northern Pacific Ocean: the pole being in the central tropical Pacific in late Precambrian time, moving across to the tropical western Pacific in the early Paleozoic and to Asia in the late Paleozoic and early Mesozoic. The data also show that the polar-wandering curve for North America is displaced westward relative to that for Europe, as Runcorn (1956b) showed, and provide an estimate for the amount of drift between the two continents since Mesozoic time, which is of the order of 30° in longitude.

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Introduction

In this paper we describe paleomagnetic observations on red beds of various geological ages in North America. This is an extension of the paleomagnetic survey described by Runcorn (1956a), in which it was established, by observations on rocks from Arizona and Utah, that the axis of the mean geomagnetic dipole (with which the axis of rotation is thought to coincide) has moved since the late Precambrian, along a roughly circular path around the northern half of the present Pacific Ocean at a rate of about $\frac{1}{3}°$ per million years. Runcorn (1956b) showed that there is a systematic difference between the polar-wandering paths inferred from American and British results, which is best interpreted by assuming that continental drift of North America westward relative to Europe took place in post-Triassic time. The considerably extended paleomagnetic survey reported in this paper supports these conclusions.

Samples of well-bedded, fine-grained red sandstones and siltstones were collected, mainly on the Colorado Plateau where no severe tectonic movements have occurred since the deposition of the sandstones. Apart from samples which fragmented during cutting or which were too weakly magnetized to be measured, the directions of magnetization of all the samples are recorded in the figures and tables of this paper. No experiments on modifying the observed magnetizations, *e.g.*, by demagnetization, are reported here, and the only correction made to the observations is that for the local geological dip at the site—the usual assumptions being made that the bedding planes were initially horizontal and that tectonic movements caused inappreciable rotation of the beds about the vertical. The measurements were carried out by the method described by Collinson, Creer, Irving and Runcorn (1957). With the astatic magnetometer used at Newcastle it has been possible to determine directions of magnetization of 10-cc specimens to accuracies within 1°–3° for intensities of magnetization as low as 3.10^{-7} emu/cc. Only about 30 discs from the whole collection had intensities too weak to be measured.

The theory of the sampling technique has been described by Runcorn (1957) and by Watson and Irving (1957), who show that a sufficiently accurate mean direction of magnetization of a rock formation (and therefore an accurate pole position) can be obtained by measurements on a comparatively small number of rock samples (10 to 20) provided these are well distributed stratigraphically. Consequently the normal procedure followed in making the collections was to sample at random intervals through the full thickness of a particular formation at one or more localities. The observed scatter of the directions of magnetization of a set of samples spanning a small fraction of a geological period is not usually greater than that to be expected from the geomagnetic secular variation (up to about 30° from the mean). Thus the random deviations between the directions of the field at the times of deposition and the present directions of magnetization of the corresponding samples may be assumed to be smaller, *i.e.*, only a few degrees. If the time spanned is of the order of a geological period, however, it is possible that superposed on the scatter due to the secular variation is that due to the movement of the axis of rotation, which may not be, over a span of a few million years, the smooth path inferred from the overall survey of the geological column. The scatter due to this cause could be of the order of 20°.

Runcorn (1956a) and Creer (1957) showed that the observations on red sandstones fall into two broad groups: (1) directions of magnetization symmetrically distributed around their mean, to which the statistical methods by Fisher (1953) may reasonably be applied, and (2) directions of magnetization scattered along a great circle through the direction of the present axial dipole field. This conclusion is again verified by the results reported in this paper. The simplest hypothesis to explain this result is to suppose that the directions of group (1) are coincident, apart from errors discussed in the previous paragraph, with the directions of the field at the time of deposition of the rock samples. It is reasonable to suppose that these rocks have been unaffected by the changing geomagnetic field since their original magnetization; they are therefore termed stable. On the other hand, group (2) appears to consist of rocks the magnetizations of which are the vector resultants of stable magnetizations acquired in the geomagnetic field at the time of deposition and secondary magnetizations acquired in the latter part of Cenozoic time in the present geomagnetic field. They are therefore termed partially stable rocks. Occasionally the effect of this secondary magnetization is so strong relative to the primary magnetization that no estimate of the latter is possible.

The directions of magnetization of lavas are

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sometimes greatly scattered (Cox, 1957; Creer, 1958) but are still symmetrical about a mean direction, from which it may be inferred that the secondary magnetization acquired by them is randomly directed. For instance, a viscous magnetization acquired during the unoriented storage of specimens between collection and measurement would be a reasonable explanation of the effects. Demagnetization by alternating fields of the order of 100 gauss removes this secondary magnetization, and the directions of magnetization become much more closely grouped. In red sandstones and siltstones no such large but symmetrical dispersion of directions has been observed. The planar distribution must therefore arise from a comparatively stable secondary magnetization lying along the direction of the present dipole field. This could be a viscous magnetization with a longer time constant than that of the lavas—the red sandstones contain hematite grains, while the lavas contain magnetite.

There are, however, other processes by which some samples of the red sandstones might have acquired this secondary magnetization. The samples collected from outcrops will have been exposed at the surface for at least some thousands of years. During this time the following processes might have occurred:

(1) A component of thermo-remanent magnetization might have been acquired through the daily heating and cooling.

(2) Water from desert storms and the heat might have caused some of the hematite grains to form iron hydroxides and carbonates and thus to lose their original magnetization. These unstable compounds might eventually form hematite again, which would then become remagnetized by the process of "chemical magnetization". It is possible that this process could occur also through the circulation of ground water and consequently affect rocks at depth.

It appears that this secondary magnetization is more frequently observed in the Western United States than in Europe (*cf.* results given by Creer, Irving, and Runcorn 1957) and could therefore be the effect of the desert climate on surface outcrops. In any case, the secondary magnetization of the red sandstones must have been acquired when the rocks were *in situ* and since the beginning of the Pleistocene, when the geomagnetic field last acquired its present polarity.

At various times mechanisms have been suggested by which the directions of magnetization of a group of samples from a rock formation could have become systematically different from the corresponding directions of the geomagnetic field at the times of their formation. The preferential settling of elongated or discoidal iron-oxide grains either horizontally or in the direction of water currents is a bias which might be introduced at the time of formation of the rock. Irreversible changes of magnetization by the process of magnetostriction caused by stress due to tectonic activity or deep burial; "chemical magnetization" occurring through the gradual change of maghemite, a common constituent of the red soils which may go to form red beds, to hematite long after deposition of the beds; or a partial thermo-remanent magnetization due to the fall of temperature after deep burial are examples of biases which could occur at any time between the formation of the rock and the present. The change in the direction of magnetization resulting from recent fields, which was discussed in the previous paragraph, is the only one of such effects for which clear evidence exists.

A further factor must be considered: If an appreciable proportion of the magnetized grains are elongated along their direction of magnetization, compaction subsequent to the magnetization process appears capable of reducing the angle of inclination. If the magnetization of the sediment occurred during deposition through the alignment of the magnetized iron-oxide grains by the geomagnetic field, an appreciable error would result only if (1) the degree of compaction were larger than, say, 5 to 10 per cent, which would not be expected in a sandstone, although possibly in a shale; and (2) the iron-oxide particles were sufficiently large to interlock with the other grains. If, on the other hand, the heavy minerals were smaller than the other grains, which is usually true in sandstones, the magnetized particles would be free to rotate, especially if the sediment were water-logged. Thus it would appear that the geomagnetic field will retain the power to orient the magnetized grains until the final stages of compaction. On the other hand chemical magnetization will occur only after the bulk of the water is squeezed out from the pores of the sediment, *i.e.*, after compaction had been substantially completed. To examine whether such effects do, in fact, bias the determinations of the mean field directions, comparisons between different formations of the same age collected in different places is necessary.

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The field work was carried out during the summers of 1955 and 1956 in co-operation with

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Dr. N. D. Opdyke, except for the Belt series which was sampled in the summer of 1957 with the help of Mr. H. R. Eisenbeis. The preparation of the rock specimens, coring, and cutting into discs was done by W. Rutherford, and the magnetic measurements and computation were made with the able assistance of A. Major, P. Dewell, and W. Rutherford in the Physics Department of King's College, Newcastle upon Tyne. This research was made possible by generous support from the Penrose Bequest of The Geological Society of America, from the Museum of Northern Arizona, Flagstaff, and from the Department of Scientific and Industrial Research, London.

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We are grateful to the Superintendents of the Grand Canyon and Glacier National Parks for permission to collect rocks in the parks.

Statistical Treatment of Paleomagnetic Directions

The statistical methods used in this paper follow the treatment of Fisher (1953), who assumes that the population from which a sample of directions of magnetization is drawn is such that the probability of a direction making an angle between θ and $\theta+d\theta$ with the true direction is proportional to $\exp(\kappa \cos \theta).d\theta$. Fisher shows that if each direction is represented by a unit vector the direction of the vector mean is the best estimate of the true direction. He also shows that a cone of confidence can be described about this mean direction which will contain the true direction at any assigned probability, the semi-angle α of the cone of confidence for a probability of 95 per cent being usually calculated. The parameter κ is a measure of the scatter, termed the precision, which is larger as the directions are more tightly grouped. For large κ the distribution becomes approximately Gaussian with a standard deviation in degrees equal to $81/\kappa^{\frac{1}{2}}$ (Runcorn, 1957).

It has been argued (*e.g.*, Blackett, 1956) that the application of such statistical methods to this subject gives an undue impression of accuracy. This point of view seems to rest on the following grounds: (1) The cone of confidence derived from the scatter of a set of observations does not include systematic errors, and it may be misleading to direct the attention of the reader to the random error in the mean direction when it may be smaller than the systematic error. (2) The analysis is based on the assumption that the directions are distributed with a frequency proportional to $\exp(\kappa \cos \theta)$, which implies assumptions about the physical cause of the scatter (Wilson, 1959). (3) Where small samples are used the divergence between the actual distribution and the theoretical one for an infinite population will be obvious by inspection.

We think that criticism (1) is mistaken because an exact statistical method is a powerful means of determining a systematic bias if one exists. This is carried out by comparing mean directions taken from two or more sets of results obtained from different localities in the same formation and testing whether the differences could arise from the finite number of samples used. The studies of the Chugwater formation in the different localities and the upper and lower parts of the Supai discussed in this paper are examples of this method of searching for systematic errors.

Objection (2) is not well founded. Runcorn (1956c) shows that if the three rectangular components of the magnetization of a series of rock samples are each distributed according to the Gauss-error function with the same standard deviations, then the directions of magnetization are distributed with a frequency proportional to $\exp(\kappa \cos \theta)$, assigning unit intensity to each sample. As the Gaussian distribution is approached where the errors are the result of a large number of independent errors, positive or negative, Fisher's distribution would appear to be appropriate to discuss the scatter of magnetization of samples, each containing a very large number of magnetized particles. Because the magnetization of the sample is weak there can be no interaction between the particles—all act independently. Further, as in Gaussian statistics, the calculations made prove to be rather insensitive to the exact form of the probability-distribution function.

Objection (3) might be raised to all modern statistical work which, having been developed largely for use in the nonphysical sciences, allows exactly for the effect of a small sample. There can be no merit in collecting larger and larger numbers of samples unless these are used effectively, *i.e.*, by an exact statistical method, and the size of the sample is simply determined by the accuracy required in the mean direction, which depends on the use to be made of the observations. The sample size used in this paper proves to be adequate in

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view of the doubt which exists of the age relationships of the rocks examined.

Conventions Used Throughout The Paper

The directions of magnetization of discs cut from samples collected from a certain section of a rock formation are plotted on a stereographic projection. Points on the lower hemisphere of the projection, *i.e.*, representing directions with positive (downward) angles of inclination, are represented by full dots. Points on the upper hemispheres of the projection, *i.e.*, representing directions with negative (upward) angles of inclination, are represented by open circles. Directions from discs cut from the same core are connected together by a full line, and directions from cores from the same rock sample are connected by a broken line. The samples are numbered from the bottom of the section upward: missing numbers represent samples either too weakly magnetized to be measured or too fragile to be satisfactorily cut to shape.

The direction which the geomagnetic field would have at the site if the earth's field were that of a dipole orientated along the present geographical axis is plotted on each stereographic projection by a square (full for sites in the present northern hemisphere).

For each set of directions at one section, the mean direction is calculated and plotted as a star. From this calculation is omitted the occasional result which is erratically displaced from the main group. The center of the star is a dot if the direction has a positive angle of inclination and an open circle if the direction has a negative angle of inclination. Described around the mean point is the 95 per cent circle or cone of confidence of radius α. Samples from some formations in which the bedding planes were easily recognizable were cored so that the axes of the cores are perpendicular to these bedding planes. The declination and inclination of the magnetization of the discs cut from these cores were thus automatically corrected for tectonic dip. The mean directions of magnetization plotted for these formations therefore can be taken as estimates of the geomagnetic field at the times of deposition. This procedure is indicated by an asterisk in the column in Table 1 headed Geological Dip. In all other cases the points on the projections give the directions of magnetization in space today, *i.e.*, the plane of the projection is the horizontal at the site. The declination and inclination given in Table 1 is corrected for local geological dip, and this corrected mean is indicated on the stereograms by a full or open triangle, according to whether the inclination is positive or negative respectively.

Where secondary magnetization is appreciable, so that the distribution of directions of magnetization is planar, the primary direction must be estimated. A point on the great circle lying near the end of the planar distribution farthest from the present dipole field is indicated by a cross on the stereograms. If the strata are flat lying this is the estimate of the primary direction. Where the rock strata are tilted and the plane of projection is the present horizontal, the planar distribution passes through the present dipole field, and the direction indicated by a cross must be corrected in the usual way for the geological dip. This point, denoted by an open or full triangle, is then the estimated primary direction of magnetization.

In Table 1, S is the number of samples collected, N denotes the number of discs cut, and N' the number used in the statistics, R is the vector sum of these N' directions (each being assigned unit length), κ the precision, and D and I the angles of declination and inclination of the vector mean direction.

Unless otherwise stated the term "pole" is always used to mean the geographical pole or pole of rotation for the geological period under discussion.

Corresponding to the 95 per cent cone of confidence for a mean direction of magnetization is a 95 per cent oval of confidence for the corresponding pole positions. In Table 1 are given the semi axes ($d\psi$, $d\chi$) of this oval along and perpendicular respectively to the great circle joining the site to the pole position.

We now proceed to describe the measurements through the geological column (Table 1).

Precambrian Formations

In the Western United States upper Precambrian (Algonkian) unmetamorphosed red beds, which have been little affected by serious tectonic movements, are prominent in the Grand Canyon series and in the Belt series.

Unkar group of the Grand Canyon series (Algonkian) (Figs. 1-3): The lowest formation of this group, exposed in the Canyon of the Colorado River, is the Bass limestone, a brown limestone which proves to have sufficient remanent magnetization for measurement. This formation was sampled near the Kaibab Trail south of the Colorado River.

Lying conformably on the Bass limestone is

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the Hakatai shale, a series of red shales, which was sampled north of the Colorado River 4.3 miles from Kaibab Bridge, just west of the Kaibab Trail, north of Phantom Ranch.

The Shinumo quartzite conformably overlies the Hakatai shale and is a hard compact fine-grained purplish quartzite. It was sampled near the Kaibab Trail south of the Colorado River.

The results show possible slight effects of recent secondary magnetization. Thus the original mean direction of magnetization of the Hakatai shale is likely to have had a more southerly declination and a smaller angle of inclination than the mean of the observed directions.

BONITO CANYON QUARTZITE (PRECAMBRIAN) (Fig. 4): The Bonito Canyon quartzite of Precambrian age was sampled east of Fort Defiance. Assuming it to have been deposited under water the well-marked ripple-marked surface was taken as the original horizontal. The quartzite rests unconformably below the Supai formation, and its age has been discussed recently by Lance (1959), who concludes that it is older than the Grand Canyon series.

BELT SERIES (ALGONKIAN) (FIGS. 5-9): The Belt series consists of a thick series of limestones, red shales, and argillites (Clapp and Deiss, 1931) which crop out mainly in Montana, where the collections reported in this paper were made. An illitic shale from the Siyeh formation of this series has recently been satisfactorily dated at 750 million years by Goldich, Baadsgaard, Edwards, and Weaver (1959).

The lowest group of the Belt series sampled was the Ravalli group in Glacier National Park, in which the Appekunny formation and the overlying Grinnell formation can be sampled easily along the road west from St. Mary's. Above these lies the Siyeh limestone at the top of which is several hundred feet of sandy and shaly beds, mostly red, which are probably correlated with the much thicker Spokane formation of the Piegan group. These were sampled at McDonald Creek. The Spokane is fully exposed in Prickly Pear Canyon, north of Helena, where it was sampled along Highway 91.

The Missoula group, possibly correlating in part with the Piegan group farther east, was sampled near Missoula, where it rests conformably on the Wallace limestone. Lowest in the section is the Miller Peak formation which was sampled in Donovan Creek, off Highway 10, east of Missoula, Montana. Above it, possibly unconformably, lies the Hellgate formation, on which the McNamara formation rests conformably. This was sampled at McNamara's Landing on State Highway 20 in Blackfoot Canyon.

LOWER PALEOZOIC FORMATIONS

Fine-grained and undisturbed red beds of Early Paleozoic age are rare in the Western United States, but the formations sampled are described.

TAPEATS SANDSTONES (CAMBRIAN) (FIG. 10): A few feet of red sandstones was found in the Tapeats sandstones in a creek bed south and west of Peach Springs, Arizona (Wood, 1956, Ph.D. Thesis, Univ. Arizona). The Tapeats sandstone can be traced almost continuously from this area northward along Peach Springs Draw to the Grand Canyon, where McKee (1945) studied the Cambrian rocks. The upper 100 feet of the Tapeats sandstone becomes finer-grained as the base of the overlying Bright Angel formation is approached, cross-bedding becomes less evident or absent, and beds of red or of yellowish-gray shaly siltstone, 6 inches to 2 feet thick, alternate with beds of light-brown to black sandstone of similar thickness. A few of the beds of the shaly siltstone are light greenish gray.

A Cambrian pole position, based on an incomplete preliminary examination of these results, was quoted by Day and Runcorn (1955) but was withdrawn in the survey of British and American paleomagnetic results by Creer, Irving, and Runcorn (1957).

BRIGHT ANGEL SHALE (CAMBRIAN) (FIG. 11): The Bright Angel shale of the Tonto group of Cambrian age of the Grand Canyon region, lying conformably below the Muav limestone and above the normally coarse-grained Tapeats sandstone, has been sampled in the Grand Wash cliffs near the Diamond Bar Ranch, Arizona, where it includes some red sandstones, as described by McKee (1945). Starting from the lower contact with the Tapeats sandstone 260 feet was sampled.

LODORE FORMATION (CAMBRIAN?) (FIG. 12): The Lodore formation was sampled along Highway 44 between Manila and Vernal, Utah, on the north flank of the Uinta Mountains, where the beds dip steeply beneath a massive Madison limestone exposure. Samples 12 and 13 are excluded from the statistical analysis. Although Williams (1953) gives the age of this formation as Cambrian, the basis for this appears to be lithological rather than paleontological.

These Cambrian results present an interesting example of secondary magnetization.

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The meaning of directions of magnetization of the Bright Angel shale and the Tapeats sandstone becomes evident when they are considered in conjunction with the Lodore formation. Although the two former are considerably scattered, the directions lie roughly about the great circle between the present dipole field and the southwest group of directions of the Lodore formation, and appear to be the result of secondary magnetizations along the present field obscuring the original directions of magnetization.

JUNIATA RED BEDS (UPPER ORDOVICIAN) (FIG. 13): The Juniata red beds of Late Ordovician age were sampled on the Pennsylvania Turnpike 2.5 miles east of Bedford, Pennsylvania, where a highway cut exposes upper Ordovician and lower Silurian sandstones in the western limb of the Friend's Cove anticline. Cores were taken perpendicular to the bedding planes of the Juniata beds, which were overturned. This exposure is much faulted, and the unusually large scatter of the results may be caused by the complex stress pattern.

Comparing these early Paleozoic pole positions and the one obtained by Graham (1949) for the Silurian, it appears that they follow a smooth path in continuation with the late Precambrian path already described. However, it would seem to be more reasonable to postulate a very late Precambrian or at least a very early Cambrian age for the Lodore formation at this locality.

UPPER PALEOZOIC FORMATIONS

DEADWOOD FORMATION (CARBONIFEROUS-MISSISSIPPIAN?) (FIG. 14): The Deadwood formation in Wyoming is usually regarded as Cambrian in age. The outcrop of the upper part of this formation in the area 35 miles northeast of Rawlins, Wyoming, which consists of red-bed deposits, has been dated by Finnell (1951, Master's Thesis, Univ. Wyoming) as probably Mississippian, from the discovery of a fragment of a bone of a member of the family Cochliodontidae. The Madison limestone of Mississippian age lies conformably above the Deadwood formation in this area. Comparison of the paleomagnetic directions of this formation and those of the Supai formation support Finnell's suggestion of a late Paleozoic age for this formation in this particular area.

NACO LIMESTONE (CARBONIFEROUS-PENNSYLVANIAN) (FIG. 15): Sandstone layers in the Naco limestone were sampled in Fossil Creek near Pine, Arizona, at a locality described by Jackson (1952).

SUPAI FORMATION (PERMIAN) (FIG. 16): The Supai formation was sampled in Oak Creek Canyon, Arizona. The lower part (members B and C) was sampled on a butte 1 mile northwest of Sedona (Fig. 16 C). The upper part (member A) was sampled up to the Fort Apache limestone (samples 30–46) above Oak Creek bridge on the west side of the canyon, and above the Fort Apache limestone (samples 47–50) on a butte near the old road from Sedona to Flagstaff (Fig. 16 B).

The Supai formation was also sampled at Hunter's Point where it is tilted on the south flank of the Fort Defiance uplift (Fig. 16 A).

There seems to be some evidence that in the thickest formations studied, such as the Supai formation in Oak Creek Canyon, there is a systematic difference in magnetization between the upper and lower parts. This is clearly shown on Figures 16 B and 16 C. We do not know, as yet, the cause of changes of magnetization within one formation or between outcrops of the same formation in different localities. However, Runcorn (1957) argues that, because the sediments of a given formation at different sites cannot be exactly contemporaneous, the discrepancies found may result from motion of the pole of rotation during the time represented by the deposits. The average movement of the pole of rotation throughout geological time is about $\frac{1}{3}°$ per million years. It is not unreasonable to suppose that for short times the pole can move at a rate of 1° per million years. Consequently, polar movements of the order of 20° can take place in a part of one geological period. Thus the change of the mean direction of magnetization through a formation may arise from this cause. Other causes of discrepancy between the mean directions of magnetization of samples collected in different parts of a formation can be suggested, but at this stage we merely wish to point out that there is a possibility that these discrepancies may represent time differences. It is known that the Defiance uplift was a positive area in the late Paleozoic, and it seems reasonable to suggest that the Supai exposures at Hunter's Point are correlated with the upper rather than the lower Supai exposures in Oak Creek Canyon. Thus the agreement between the mean direction in Figures 16 A and B appears to support the above argument.

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Triassic Formations

MOENKOPI FORMATION (EARLY AND MEDIAL TRIASSIC AGE) (FIG. 17): The Moenkopi formation of the Colorado Plateau consists of interfingered marine and continental beds. In Arizona and Utah the continental beds predominate; they consist of red claystones, siltstones, and sandstones, which accumulated on mud flats and deltas. The most recent monograph on this formation is by McKee (1954).

The Moenkopi was sampled at the following sites: (A) Capitol Reef National Monument, Utah; (B) Sand Canyon, Dinosaur National Monument, Colorado; (C) Split Mountain, Dinosaur National Monument, Colorado; (D) Vernal, Utah; (E) Poverty Tank, 5 miles west of Cameron, Arizona; (F) Echo cliffs, 8 miles south of the Navajo Bridge, Arizona (section 10 of Figure 5 and section 8 of Figure 6 respectively of the northwest-southeast series of sections of the Moenkopi formation given by McKee, 1954).

CHUGWATER FORMATION (TRIASSIC) (FIG. 18): This formation was sampled from the following sites in Wyoming: (A) Red Mountain, Albany County; (B) Rawlins, Carbon County; (C) east of Thermopolis, Hot Springs County; (D) north of Lander, Fremont County; (E) Dinwoody Lakes in Wind River Range, Fremont County; (F) Fort Washakie in Wind River Range, Fremont County; (G) a butte north of Shell, Big Horn County; (H) Sheep Mountain, north of Greybull, Bighorn County; (I) north of Alcova Reservoir, Natrona County; (J) Troublesome Creek, Carbon County.

For the sections where opposed directions of magnetization were found, the means of the two mean directions from the opposite groups were used to calculate the pole. This procedure should eliminate small components of secondary magnetization. Thus on the corresponding figures the great circle has been drawn through the present dipole field direction and this final mean axis, the circles terminating at the latter.

These results are in agreement with the angles of declination determined from seven bore-hole cores from the Chugwater formation near Speas dome and east of Alcova, which lay between 4° and 48° east of south. The orientation of the bore-hole cores was obtainable in this case because the beds dipped in a direction known from the general geological structure in the region. This information has been kindly made available by the Pan-American Oil Company of Tulsa, Oklahoma.

Figures 19 and 20 show the mean directions of magnetization of the various opposed groups of the Chugwater formation at the different localities before and after correction for local geological dip. It will observed that the points are better grouped after correction. This, of course, is an argument in favor of the hypothesis that, apart from a small component of viscous magnetization along the present dipole field, the magnetization is an original one. In many cases the circles of confidence do not overlap. There is therefore some suggestion of systematic difference between the directions of magnetization of the Chugwater formation at different localities which, as in the case of the Supai formation discussed above, could result from age differences between the sections.

NEWARK GROUP (UPPER TRIASSIC) (FIG. 21): An exposure of the basal Brunswick formation of the Newark series of the Triassic red beds of eastern North America (McLaughlin, 1950) was sampled in the Delaware River valley south of Frenchtown, New Jersey. These results have been discussed by Du Bois, Irving, Opdyke, Runcorn, and Banks (1957).

CHINLE FORMATION (UPPER TRIASSIC) (FIG. 22): The Chinle formation of Late Triassic age described by Harshbarger, Repenning, and Irwin (1957) was sampled in the Moab area and in the Colorado National Monument at the following sites: (A) Colorado National Monument, (B) Colorado National Monument, (C) along the highway north of Moab, (D) east of Moab on the south side of the Colorado River. It will be seen that the results at sites A and B are more consistent with a Jurassic than with a Triassic age.

Jurassic and Tertiary Formations

In these formations directions of magnetization are found which are little different from those of the present dipole field.

KAYENTA FORMATION (LOWER JURASSIC) (FIG. 23): The Kayenta formation has been described by Harshbarger, Repenning, and Irwin (1957). Collections were made at the following sites: (A) East of Kanab, Arizona; (B) in the Navajo National Monument, Arizona; (C) in the Echo Cliffs at the Gap, Arizona; (D) the type locality north of Kayenta, Arizona.

CARMEL FORMATION (JURASSIC) (FIG. 24): The Carmel formation was sampled south of Moab, Utah.

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TERTIARY RED BEDS IN UTAH (FIG. 25): Tertiary red beds in the Duchesne River formation were sampled near Roosevelt, Utah. A few samples of gray marine sandstones of Pliocene age from the eastern Los Angeles basin, California, were measured. The local geological dip of the beds was between 45° and 55°. Their directions of magnetization in space at the present time were found to be close to that of the present geomagnetic field. (The mean was N. 17° W. + 51°; *cf.* present dipole field N. 0° + 53°.) These rocks were probably tilted in the last million years, so that these sandstones have a negligible stable component of magnetization. Gray, white, and green sandstones often seem to have a weak and unstable magnetization in contrast to the red sandstones on which paleomagnetic surveys have usually been based. Miller and Folk (1955) have noticed that red sandstones show a relative abundance of iron oxide compared to white, gray, and green sandstones. This fact alone probably accounts for the much stronger magnetization found in red sandstones. The greater stability of the magnetization of the red sandstones is probably due to the presence of hematite which has a much higher coercive force than does the magnetite present in the white, gray, and green sandstones.

CONCLUSIONS

Polar-Wandering Curves

Figures 26 and 27 show the positions of the poles calculated by the authors quoted above, by Runcorn (1956a), and from the data presented in this paper. Where it is clear that the distribution is substantially free from secondary magnetization, a pole position is calculated directly from the vector mean direction.

The presence of secondary magnetization obscures the original directions of magnetization to a varying degree. However, even if the secondary magnetization has an extremely important effect, as in the Chinle formation, the great circles for rocks of the same age coincide. Runcorn (1956b) shows that for flat-lying rocks it is possible to determine exactly from the position of the great circle the longitude of the pole position at that geological time. Its latitude, however, in the absence of an experimental method of removing the secondary magnetization, may only be roughly estimated. Whether the rock strata are flat lying or not, an estimated pole is calculated: if flat lying, from the point on the great circle near the end of the distribution farthest from the present dipole field, indicated by a cross on the stereograms; if tilted, from this estimated direction after correction for geological dip, indicated by a triangle on the stereograms. The corresponding pole position is then plotted.

Figure 26 gives the mean pole positions for different formations except that the Hakatai shale poles for the two different sites are plotted. It also shows that the poles from successively older geological periods, with the exception of the Mississippian poles, lie progressively farther from the present pole along a reasonably smooth path. We consider this to be a most significant fact, from which the inference may be drawn that the paleomagnetic directions record a phenomenon which changes gradually with geological time. It is therefore more reasonable to explain it primarily by a geophysical change affecting the earth as a whole rather than by the local geological and mineralogical processes, which were discussed in the introduction to this paper. It is therefore concluded that the major part of the magnetization of these red beds was acquired parallel to the geomagnetic field about the time of their deposition.

The pole positions of the upper Precambrian formations are grouped together near the equator in the central Pacific (Fig. 26). There is some evidence that they are spread in the overall direction of the polar-wandering curve, and the differences could reflect the long period of time represented by these rocks. The age of the Siyeh formation quoted above indicates that this period could be 100-200 million years, and, if the overall rate of polar wandering is taken as about 0.3° per million years, then the observed spread in the pole positions is explicable. The inference that the Grand Canyon series and the Belt series are contemporaneous is in agreement with the conclusions of geologists who have studied these formations. The considerable divergence between the pole positions determined by Runcorn (1956a) and Doell (1955) for the Hakatai shales on the south side of the Colorado River, which are in good agreement, and those determined in this paper for the other formations in the Grand Canyon series at the same site and the Hakatai shales on the north side of the Colorado River is puzzling. The discrepancy in latitude can be reduced by assuming a component of secondary magnetization. If this were allowed for, the north magnetic pole would be moved southward (Runcorn, 1956b).

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The directions of magnetization of samples from the Grinnell formation (Fig. 6) and the Spokane formation from Prickly Pear Canyon (Fig. 8) are distributed with a slight spread in the direction of the present dipole field. This suggests that their vector means require a small correction for secondary magnetization in the latter direction. In Figure 26 arrows are drawn from pole positions A.6 and A.8, calculated from the vector mean directions in Figure 6 and Figure 8, respectively, to pole positions, corrected by the usual method for these effects. All the late Precambrian pole positions then come very close together, except for the Hakatai shales on the south rim of the Canyon.

Figure 26 shows that the poles for the Belt series and for the Grand Canyon series lie on the polar-wandering curve in order of their stratigraphical positions, except for that of the McNamara formation (pole A.9b), which, stratigraphically, is the youngest of the Belt rocks and might, therefore, be expected to lie west of the other poles instead of to the east. The question of whether the sampled section of this formation is really older than the sampled sections of the other formations of the Belt series in spite of its apparent stratigraphic position or whether in fact the pole moved back on its path during these times must be left open. The similar anomaly of pole A.7 is not, perhaps, of significance in view of the small number of samples available.

The Bonito Canyon quartzite has a pole near those of the Belt and Grand Canyon series. This result is therefore evidence against the older Precambrian age that Lance (1959) suggests, unless the Bonito Canyon quartzite acquired its present magnetization long after its formation, possibly as a result of slight metamorphism. Allowance for a possible small component of secondary magnetization would move the south magnetic pole computed for this formation (which is the pole plotted on Fig. 26) northward, closer to the main group of the late Precambrian poles.

So far it has been assumed that the directions of magnetization of a series of samples taken from a certain rock series are scattered randomly about a mean direction, which corresponds to a definite pole position. It is necessary now to consider what modification arises if the rock series spans a sufficient time for a small motion of the pole to take place, which to a first approximation can be regarded as occurring along an element of a great circle. Suppose we now assume that a number of sets of samples are collected from the rock series, but that each set is differently distributed in time. If the number of measurements in each set is sufficient to reduce to negligible amounts the effect of the random scatter on the mean directions from each set of samples, then the pole positions calculated from the mean directions will not coincide but will be scattered along the path of the pole during the time considered. In the sampling of a rock series cropping out in different places, it is inevitable that collections of samples from different localities will be weighted differently with respect to the time spanned by the whole rock series. But it is clear that the mean pole position determined from the pole positions calculated from the different localities will tend to approximate the pole position corresponding to the mean of the time interval covered by the entire rock formation.

Figure 27 shows the pole positions determined by the authors quoted above, by Runcorn (1956a), and in this paper from the various localities for the Carboniferous, Permian, and Triassic. The scatter of the Triassic poles appears to be slightly greater in the general direction of the polar movement and may possibly be partly due to time differences between the various sections. Pole positions for the two Chinle localities A and B are not shown because they lie close to the present pole.

In this paper the circles of confidence of the mean directions of magnetization have been calculated giving each disc unit weight. Whether this is entirely desirable depends on which of the possible causes of dispersion of directions of magnetization is most important in a particular set of observations. Dispersion due to the geomagnetic secular variation is likely to be present in most of the collections discussed in this paper, for Runcorn (1957) has shown, by order-of-magnitude arguments, that discs of sandstone were deposited in times short compared with the time scale of the geomagnetic secular variation. However, this is unlikely to be true in limestones in which the rate of deposition may be comparable with that of ocean sediments, *i.e.*. 1 mm/1000 years. Confirmation of this is given by the directions of magnetization of the Bass limestone (Fig. 1). They are slightly distributed along a great circle due to secondary magnetization along the present dipole field, but they are, however, scattered through only a few degrees perpendicular to the great circle.

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The errors of collection, cutting, and measurement are of the order of 1°–3°. Consequently, it appears that in the limestone the geomagnetic secular variation is smoothed out even in a single disc.

Therefore unless many pairs of sandstone discs are exactly contemporaneous (to within a few years) the procedure followed in calculating the circles of confidence of the mean direction of magnetization and the ovals of confidence of the pole positions appears to be realistic.

The polar-wandering path from North American rocks given by Runcorn (1956a) and the one which we suggest on the basis of the new information do not differ significantly. The one determined from the data in this paper lies farther south in the early Paleozoic and Precambrian.

It is interesting to determine from Figure 26 the rate of polar movement relative to North America. Table 3 gives the angular distance between successive poles, the approximate time interval between them, using the time scale given by Holmes (1944, p. 105), and the calculated rate of polar movement. The rate is roughly constant, the mean rate being about 0.3° per million years in agreement with that given by Creer, Irving, and Runcorn (1957).

Comparisons with Results of Other Workers

The results described in this paper are consistent with those obtained by most other workers.

Howell and Martinez (1956; 1957) sampled concretions in the Barnett formation of Mississippian age in the Llano uplift in Texas. From seven localities with southeast magnetization and one locality with a northwest magnetization they estimated the pole positions at 41° N., 128° E., and 42° N., 142° E. respectively. From the Point Peak member of the Wilberns formation in the same area Howell and Martinez (1957) also obtained a Cambrian pole position at 0° N., 158° E. In each of these formations secondary magnetizations were found, and the quoted results roughly allow for this. These results agree quite well with those given by Runcorn (1956a) and in this paper.

Doell (1955) sampled all the major sedimentary formations of the Grand Canyon but found that only the Hakatai shales and Supai shales are strongly and consistently magnetized. For the Hakatai shales it is necessary to allow for the small local geological dip (Creer, Irving, and Runcorn, 1957). Doell's Algonkian pole position is at 13° N., 127° W., and his Permian pole at 39° N., 115° E., which agree reasonably well with those given by Runcorn (1956a)—31° N., 150° W., and 26° N., 121° E. respectively, from nearly the same sites. Doell states that his Precambrian results were badly scattered, the angle of the circle of confidence being 18°. The Supai results obtained by Runcorn (1956a) show some effect of secondary magnetization, but the pole position given was not corrected to allow for this, *i.e.*, it was simply computed from the vector mean direction. If rough allowance is made for this the direction of magnetization is given by D = S. 40° E. and I = +12°, corresponding to a pole position at 33° N., 120° E.

Du Bois (1957) has determined poles for the Keweenawan rocks around Lake Superior. Those for the Freda and Nonesuch shales, the magnetic stability of which can be demonstrated, and the Jacobsville sandstone (9°N. 169°E., 14°S. 170°W., respectively) fall close to the main group of the Grand Canyon series. The poles for the Portage Lake and Copper Harbor formation (25°N. 170°W., and 30°N 176°E., respectively) are also in rough agreement with the results of this paper on the Precambrian but are closer in longitude than in latitude. However, the pole of the Chequamegon (69°S. 133°E.), which is the stratigraphically highest formation in the Keweenawan series, has such a high latitude that it is possible that it has a high degree of secondary magnetization.

Thus, apart from some instances of secondary magnetization along the present dipole field direction, the mean directions of magnetization of red sandstones of the same geological age from different sites give concordant results; the discrepancies are of the order which could result from differences in age. This result conflicts with the conclusion implied by some results of Kintzinger (1957), who determined the magnetization of samples of the Moenkopi and the Dinosaur Canyon member of the Moenave formation near Marble Canyon, Arizona. He found a direction of magnetization for the Moenkopi of D = N. 35° W. and I = +35° in reasonable accord with the results given above. Runcorn (1956a) gives a determination of the direction of magnetization of the Moenkopi formation which is significantly different from those of this paper and from Kintzinger's determination, but he stated that the great scatter and the inhomogeneous

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character of the particular strata sampled made the determination of dubious accuracy. However, Kintzinger's results on the Dinosaur Canyon member and those of Graham (1955) on the Chinle from New Mexico, which he quotes, give a declination well to the east of north with a positive angle of inclination, for which there is no parallel in Triassic rocks of the Western United States. The cause of this anomaly is not known, although these results of Kintzinger and Graham are based on very few samples.

Evidence for Continental Drift

Runcorn (1956b) shows that the polar-wandering path calculated from North American results lies to the west of that calculated from Europe. This conclusion is confirmed by the results of this paper (Fig. 26).

He argued that the discrepancy between the polar-wandering paths for North America and Europe could not be explained as a systematic difference in the amount of secondary magnetization in the British and American rocks, because this component would not affect the longitude of the pole positions, and therefore a relative shift of North America westward of about 24° with respect to Europe since Mesozoic time was postulated. Table 2 shows that the differences of longitude between the British and American poles, calculated from the results in this paper, are greater than the 24° determined from the earlier data by Runcorn (1956b) and closer to that suggested by Du Bois (1957). If, prior to middle Mesozoic time, the continents of Europe and North America were very much nearer and possibly in contact the discrepancy between the two polar-wandering paths would be negligible. But these paths are similar in shape only to a first approximation. In particular the kinks in both curves at the end of the Paleozoic seem real and may be associated with the beginning of the postulated continental displacement of North America westward relative to Europe. To infer, in detail, the relative movements of the two continents through geological time we must await more detailed paleomagnetic surveys in each continent. Even then the fact that three angles are required to specify the position of a continent relative to the earth's axis at any time, whereas paleomagnetic observations supply only two, gives rise to an indeterminacy that must be resolved by geological considerations.

Evidence Relating to Reversals of the Geomagnetic Field

The following explanations of the opposed directions of magnetization found in many formations have been suggested:

(1) At irregular intervals in the earth's history, of the order of millions of years, the geomagnetic field reversed its polarity as a result of changes in the magnetohydrodynamic interactions in the fluid core of the earth. The actual reversal of the field would occupy a short time (of the order of thousands of years), and during this interval the axial symmetry of the field about the earth's axis of rotation would be lost.

(2) Because of the complex mineralogy of the iron oxides they contain, certain rocks have the property of becoming magnetized antiparallel to the geomagnetic field at the time of their formation.

Were the alternations between opposed directions of magnetization sufficiently well spread in the geological column, it would be a simple matter to test whether or not reversals of the geomagnetic field have really occurred by correlation over large distances, for if (2) happens frequently we should not find the same sign of magnetization at different places in rocks of the same age. The following considerations from the present study are relevant to this discussion:

(1) So far extensive measurements of rocks of Permian age in North America and Europe have given magnetization of one sign only (positive, *i.e.*, the south magnetic pole near Asia on the convention of Creer, Irving, and Runcorn).

(2) In Triassic time both European and American rocks show more than one reversal of magnetization. Table 4 shows the sequence of reversals of magnetization at the various sites of the Chugwater and Moenkopi formations. The western exposures of the Chugwater formation in Wyoming all show three or four zones of opposed magnetization which may correspond. On the other hand each of the eastern and northern exposures shows only one zone, in each case positive (except for the single negative sample at Shell). This suggests that the time spanned by the western exposures is longer than that by the eastern exposures. Red Mountain section in the south was among the most extensively sampled sections and shows four zones. The Utah exposures of the Moenkopi formation are all of positive sign,

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except for the single uppermost sample in Sand Canyon, and the Arizona exposures show three or four successive reversals. Thus it can be concluded that closely related sections which are likely to cover similar time intervals show the same number of alternations of magnetization.

This evidence from rather widely scattered localities appears to us to be easier to reconcile with explanation (1) than with (2).

References Cited

Blackett, P. M. S., 1956, Lectures on rock magnetism: Jerusalem, Weizmann Science Press, 131 p.

Clapp, C. H., and Deiss, C. F., 1931, Correlation of Montana Algonkian formations: Geol. Soc. America Bull., v. 42, p. 673–696

Collinson, D. W., Creer, K. M., Irving, E., and Runcorn, S. K., 1957, The measurement of the permanent magnetization of rocks: Royal Soc. Philos. Trans. A, v. 250, p. 73–82

Cox, A., 1957, Remanent magnetization of lower to middle Eocene basalt flows from Oregon: Nature, v. 179, p. 685–686

Creer, K. M., 1957, The remanent magnetization of unstable Keuper Marls: Royal Soc. Philos. Trans. A, v. 250, p. 130–143

——1958, Preliminary paleomagnetic measurements from South Africa: Ann. de Geophys., v. 14, p. 373–390

Creer, K. M., Irving, E. and Runcorn, S. K., 1957, Geophysical interpretation of paleomagnetic directions from Great Britain: Royal Soc. Philos. Trans. A, v. 250, p. 144–156

Day, A. A., and Runcorn, S. K., 1955, Polar Wandering—some geological, dynamical and paleomagnetic aspects: Nature, v. 176, p. 422–426

Doell, R. R., 1955, Paleomagnetic study of rocks from the Grand Canyon of the Colorado River: Nature v. 176, p. 1167

Du Bois, P. M., 1957, Comparison of paleomagnetic results for selected rocks of Great Britain and North America: Adv. in Physics, v. 6, p. 177–186

Du Bois, P. M., Irving, E., Opdyke, N. D., Runcorn, S. K., and Banks, M. R., 1957, The geomagnetic field in Upper Triassic times in the United States: Nature, v. 180, p. 1186–1187

Fisher, R. A., 1953, Dispersion on a sphere: Royal Soc. Philos. Proc. A, v. 217, p. 295

Goldich, S. S., Baadsgaard, H., Edwards, G. and Weaver, C. E., 1959, Investigations in radioactivity—dating of sediments: Am. Assoc. Petroleum Geologists Bull. v. 43, p. 654–662.

Graham, J. W., 1949, The stability and significance of magnetism in sedimentary rocks; Jour. Geophys. Research, v. 54, p. 131–167

——1955, Evidence of polar shift since Triassic times: Jour. Geophys. Research, v. 60, p. 329–347

Harshbarger, J. W., Repenning, C. A. and Irwin, J. H., 1957, Stratigraphy of the uppermost Triassic and the Jurassic rocks of the Navajo Country: U. S. Geol. Survey Prof. Paper 291

Holmes, A., 1944, Principles of physical geology: London, Nelson, 532 p.

Howell, L. G., and Martinez, J. D., 1956, Paleomagnetism of chemical sediments: Nature, v. 178, p. 204

——1957, Polar wandering as indicated by rock magnetism: Geophysics, v. 22, p. 384–397

Jackson, R. L., 1952, Pennsylvanian-Permian facies of the Supai formation in central Arizona: Ariz. Geol. Soc. Guidebook, S. Arizona Excursion

Kintzinger, P. R., 1957, Paleomagnetic survey of Triassic rocks from Arizona: Geol. Soc. America Bull., v. 68, p. 931–932

Lance, J. F., 1959, Precambrian rocks of northern Arizona: New Mexico Geol. Soc. 9th Field Conf. Handbook, p. 66–70

McKee, E. D., 1945, Cambrian history of the Grand Canyon region: Carnegie Inst. Washington Pub. 563, pt I

——1954, Stratigraphy and history of the Moenkopi formation of Triassic age: Geol. Soc. America Mem. 61, 133 p.

Miller, D. N., and Folk, R. L., 1955, Occurrence of detrital magnetite and ilmenite in red sediments: new approach to singificance of red beds: Am. Assoc. Petroleum Geologists Bull., v. 39, p. 338–345

McLaughlin, D. B., 1950, A suggested correlation of Triassic areas of the eastern United States: Pa. Acad. Sci. Proc., v. 24, p. 161–169

Nairn, A. E. M., 1960, Paleomagnetic results from Europe: Jour. Geology, v. 68, p. 285–306

Runcorn, S. K., 1956a, Paleomagnetic survey in Arizona and Utah; Preliminary results: Geol. Soc. America Bull., v. 67, p. 301–316

——1956b, Paleomagnetic comparisons between Europe and North America: Geol. Assoc. Canada Proc., v. 8, p. 77–85

——1956c, Field observations of rock magnetism: Intern. Geol. Cong. Proc., Mexico City

——1957, The sampling of rocks for paleomagnetic comparison between the continents: Adv. in Physics, v. 6, p. 169–176

Watson, G. S., and Irving, E., 1957, Statistical methods in rock magnetism: Royal Astron. Soc., Mon. Not. Geophys. Suppl., v. 7, p. 290–299

Williams, N. C., 1953, Late Precambian and early Paleozoic geology of Western Uinta Mountains, Utah: Am. Assoc. Petroleum Geologists Bull., v. 37, p. 2734–2742

Wilson, R. L., 1959, Remanent magnetism of late Secondary and early Tertiary British rocks: Philos. Mag., v. 4, p. 750–755

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TABLE 1.—DIRECTIONS OF MAGNETIZATION, STATISTICS, AND POLE POSITIONS

Fig. No.	Formation	Geol. Age	Locality	Township and Range	Lat.	Long.	Thickness sampled (ft.)	Geol. Dip										Pole position			
								Str.	Dip	S	N	N′	R	D	I	α	κ	Lat.	Long.	$d\psi$	$d\chi$
25	Duchesne River	T	Roosevelt	S22, T. 6 S., R. 1 W.	40° N.	110° W.	50	—	—	24	85	85	78.9	N. 2° E.	+65°	5°	13.8	83° N.	99° W.	6°	8°
24	Carmel	J	Moab	S22, T. 27 S., R. 22 E.	39° N.	109° W.	50	—	—	9	31	31	28.0	N. 11° W.	+63°	9°	10.0	80° N.	160° W.	11°	14°
23 (A)	Kayenta	J	Kanab	S33, T. 43 S., R. 7 W.	37° N.	113° W.	100	—	—	10	39	33	30.15	N. 4° E.	+53°	9°	11.6	85° N.	23° E.	9°	13°
23 (B)	Kayenta	J	Navajo Nat. Mon.	S20, T. 38 N., R. 14 E.	37° N.	111° W.	200	—	—	7	27	27	23.2	N. 25° W.	+50°	11°	6.9	—	—	—	—
23 (C)	Kayenta	J	Echo Cliffs	S32, T. 34 N., R. 9 E.	37° N.	112° W.	140	—	—	14	43	43	40.8	N. 14° E.	+43°	6°	13.7	73° N.	19° E.	5°	7°
23 (D)	Kayenta	J	Kayenta	S2, T. 38 N., R. 19 E.	37° N.	110° W.	50	N. 49° E.	16° S.	8	43	43	38.9	N. 20° E.	+50°	6°	13.5	72° N.	6° W.	6°	8°
22 (B)	Chinle	Tr	Colorado Nat. Mon.		39° N.	109° W.	50	N. 60° W.	3° NE.	7	31	31	28.79	N. 34° E.	+60°	7°	13.6	—	—	—	—
22 (A)	Chinle	Tr	Colorado Nat. Mon.		39° N.	109° W.	100	—	—	6	29	29	27.96	N. 4° W.	+66°	5°	25.4	—	—	—	—
22 (D)	Chinle	Tr	Moab	S20, T. 25 S., R. 22 W.	39° N.	109° W.	140	—	—	10	60	60	20.48	N. 59° E. S. 20° E.	+73° −10°	—	—	48° S. 50° N.	114° E. 114° E.	—	—
22 (C)	Chinle	Tr	Moab	S20, T. 25 S., R. 22 W.	39° N.	109° W.	220	N. 46° W.	20° SW.	14	39	39	25.56	S. 20° E. S. 24° E.	+44° −7°	—	—	23° S. 49° N.	90° E. 109° E.	—	—
21	Newark	Tr	Frenchtown		41° N	75° W	100	N. 45° E.	10° NW.*	20	78	70	69.19	N 6° E	+28°	3°	86.3	64° N	92° E.	2°	3°
18 (A)	Chugwater	Tr	Red Mtn.	S15, T. 12 N., R. 76 W.	41° N.	106° W.	160	N. 73° E.	13° S.	18	75	16	15.48	N. 25° W.	+15°	6°	30.0	47° N.	116° E.	3°	5°
												52	47.72	S. 29° E.	−6°	6°	11.7				
18 (B)	Chugwater	Tr	Rawlins	S22, T. 23 N., R. 88 W.	42° N.	107° W.	90	N. 66° W.	8° NE.	9	33	12	11.75	N. 32° W.	+12°	4°	43.1	46° N.	121° E.	3°	5°
												21	20.22	S. 31° E.	−19°	6°	25.3				
18 (C)	Chugwater	Tr	Thermopolis	S21, T. 42 N., R. 94 W.	44° N.	108° W.	400	N. 75° W.	10° NE.	6	33	18	17.68	N. 28° W.	+16°	5°	56.0	52° N.	103° E.	4°	7°
												15	13.94	S. 10° E.	−19°	11°	13.2				
18 (D)	Chugwater	Tr	Lander	S4, T. 31 N., R. 99 W.	43° N.	109° W.	320	N. 43° W.	18° NE.	10	49	12	11.89	N. 16° W.	+27°	4°	55.0	47° N.	100° E.	4°	8°
												37	32.34	S. 23° E.	+15°	9°	7.7				
18 (E)	Chugwater	Tr	Dinwoody Lakes	S24, T. 5 N., R. 6 W.	43° N.	109° W.	220	N. 45° W.	15° NE.	10	42	20	19.37	N. 21° W.	+25°	6°	30.2	52° N.	111° E.	3°	5°
												22	20.76	S. 26° E.	−20°	7°	17.0				
18 (F)	Chugwater	Tr	Fort Washakie	S26, T. 15, R. 2 W.	43° N.	109° W.	120	N. 39° W.	11° NE.	11	53	25	23.94	N. 14° W.	+14°	6°	22.5	44° N.	105° E.	3°	5°
												28	27.20	S. 34° E.	+6°	5°	33.6				
18 (G)	Chugwater	Tr	Shell	S24, T. 53 N., R. 90 W.	45° N.	108° W.	100	—	—	5	21	5	—	N. 20° W.	+36°	—	—	—	—		
												16	14.99	S. 25° E.	−23°	7°	16.0	56° N.	113° E.	4°	8°
18 (H)	Chugwater	Tr	Sheep Mtn.	S2, T. 53 N., R. 94 W.	45° N.	108° W.	70	N. 52° W.	20° SW.	9	38	35	33.43	S. 32° E.	−6°	3°	58.0	40° N.	115° E.	2°	3°
18 (I)	Chugwater	Tr	Alcova Res'vr.	S14, T. 30 N., R. 83 W.	43° N.	107° W.	90	N. 36° W.	11° E.	10	48	48	34.2	S. 28° E.	−18°	8°	9.75	49° N.	118° E.	4°	8°
18 (J)	Chugwater	Tr	Troublesome Creek	S20, T. 25 N., R. 81 W.	42° N.	107° W.	90	N. 68° E.	28° S.	9	43	43	38.71	S. 46° E.	−12°	7°	10.0	36° N.	135° E.	4°	7°

17 (A)	Moenkopi	$\overline{\text{R}}$	Capitol Reef	S16, T. 29 N., R. 6 E.	38° N.	111° W.	70	—	—	10	39	39	30.65	S. 41° E.	+24°	—	—	26° N.	114° E.	—	—
														S. 29° E.	−6°			46° N.	114° E.		
17 (B)	Moenkopi	$\overline{\text{R}}$	Sand Canyon	S10, T. 15, R. 7 E.	40° N.	109° W.	100	—	—	8	30	27	24.48	S. 32° E.	−7°	9°	10.5	43° N.	118° E.	5°	9°
17 (C)	Moenkopi	$\overline{\text{R}}$	Split Mtn.		40° N.	109° W.	100	—	—	5	23	23	22.20	S. 24° E.	−4°	6°	27.5	46° N.	105° E.	3°	6°
17 (D)	Moenkopi	$\overline{\text{R}}$	Vernal	S35, T. 25, R. 21 E.	41° N.	109° W.	50	N. 39° E.	15° E.	9	41	38	35.96	S. 22° E.	−4°	6°	17.7	46° N.	104° E.	3°	6°
17 (E)	Moenkopi	$\overline{\text{R}}$	Poverty Tank	S21, T. 29 N., R. 9 E.	36° N.	111° W.	120	N. 65° E.	4° N.	11	30	27	25.80	N. 23° W.	+36°	7°	21.7	64° N.	127° E.	5°	8°
17 (F)	Moenkopi	$\overline{\text{R}}$	Echo Cliffs	S2, T. 39 N., R. 7 E.	37° N.	112° W.	320	—	—	22	48	42	39.53	N. 11° W.	+28°	6°	16.6	66° N.	95° E.	4°	6°
16 (A)	Supai	P	Hunter's Point	S2, T. 27 N., R. 6 W.	36° N.	109° W.	100	N. 3° W.	25° E.	7	24	24	23.29	S. 16° E.	+5°	5°	32.0	49° N.	96° E.	3°	5°
16 (B)	Supai	P	Oak Creek Canyon	S1, T. 18 N., R. 6 E.	35° N.	112° W.	480	—	—	14	32	32	29.54	S. 19° E.	+10°	8°	12.6	46° N.	96° E.	4°	8°
16 (C)	Supai	P	Oak Creek Canyon	S5, T. 17 N., R. 6 E.	35° N.	112° W.	350	—	—	16	57	55	52.86	S. 39° E.	+18°	4°	25.0	33° N.	116° E.	2°	4°
15	Naco limestone	C_P	Fossil Creek	S29, T. 12 N., R. 7 E.	34° N.	112° W.	150	—	—	9	20	20	19.12	S. 55° E.	+16°	7°	21.6	23° N.	130° E.	4°	7°
14	Deadwood	C_M	Rawlins	S17, T. 21 N., R. 87 W.	42° N.	107° W.	50	S. 46° W.	12° SE.	7	34	34	31.88	S. 29° E.	−14°	7°	15.5	47° N.	117° E.	4°	7°
13	Juniata	O	Bedford		40° N.	79° W.	1500	N. 39 E.	75° SE.*	12	60	56	47.18	S. 49° E.	+26°	8°	6.2	20° N.	153° E.	5°	9°
12	Lodore	ϵ	Vernal	S18, T. 6 N., R. 19 E.	41° N.	110° W.	200	S. 88° W.	28° N.*	11	38	26	24.20	N. 59° E.	+4°	8°	13.9	23° S.	174° W.	4°	7°
												7	6.76	S. 54° W.	+13°	13°	25.0				
11	Bright Angel Shales	ϵ	Diamond Bar Ranch	S8, T. 28 N., R. 16 W.	36° N.	114° W.	260	—	—	20	44	—	—	—	—	—	—	—	—	—	—
10	Tapeats	ϵ	Peach Spring	S3, T. 25 N., R. 11 W.	36° N.	114° W.	10	—	— —	15	47	—	—	—	—	—	—	—	—	—	—
9b	McNamara	A	Blackfoot canyon	S10, T. 13 N., R. 17 W.	47° N.	114° W.	84	N. 75° W.	22° N.*	20	62	53	51.29	N. 26° E.	−43°	4°	30.0	14° S.	138° W.	3°	5°
9a	Miller Peak	A	Donovan Creek	S26, T. 13 N., R. 18 W.	47° N.	114° W.	65	N. 35° W.	41° SW.*	14	23	23	21.90	S. 54° W.	+30°	7°	20.0	11° S.	166° W.,	4°	8°
8	Spokane	A	Prickly Pear Canyon	S3, T. 13 W., R. 4 W.	47° N.	112° W.	400	N. 68° E.	10° NW.	39	72	71	67.16	S. 52° W.	+55°	4°	18.2	5° N.	152° W.	4°	6°
7	Spokane	A	McDonald Creek		49° N.	114° W.	50	N. 46° W.	19° NE.	5	19	19	17.92	S. 26° W.	+39°	8°	9.9	16° S.	139° W.	6°	10°
6	Grinnell	A	Glacier Park		49° N.	114° W.	70	N. 44° W.	16° SW.*	16	44	44	41.1	S. 45° W.	+48°	6°	14.8	2° S.	152° W.	5°	8°
5	Appekunny	A	Glacier Park		49° N.	114° W.	110	N. 38° W.	16° SW.	15	39	38	36.40	S. 43° W.	+29°	6°	14.6	15° S.	156° W.	4°	7°
4	Bonito Canyon quartzite	A	Fort Defiance, Ariz.		36° N.	109° W.	100	N. 15° W.	20° SW.*	16	75	74	70.10	N. 31° E.	−25°	4°	18.7	33° S.	146° W.	2°	4°
3	Shinumo quartzite	A	Grand Canyon		36° N.	112° W.	100	N. 28° E.	10° SE.	14	61	61	50.65	N. 72° W.	+65°	—	—	37° N.	166° W.	—	—
														S. 66° W.	+33°			7° S.	173° W.		
2	Hakatai shales	A	Grand Canyon		36° N.	112° W.	36	N. 57° W.	8° NE.	14	44	41	37.82	N. 69° W.	+58°	—	—	36° N.	177° W.	—	—
														S. 65° W.	+31°			9° S.	174° W.		
1	Bass limestone	A	Grand Canyon		36° N.	112° W.	300	N. 28° E.	10° SE.	14	45	43	41.19	S. 52° W.	+52°	5°	23.2	6° S.	154° W.	3°	6°
														S. 45° W.	+34°			21° S.	158° W.		

S—Number of samples collected; N—Number of discs measured; N′—Number of discs used in the determination of the mean direction; R—Vector sum of the N′ directions (each assigned unit length); D—Mean declination; I—Mean inclination; α—Semi-angle of cone of confidence; κ—Precision (see text); $d\psi$, $d\chi$—Semi-axes of the oval of confidence around each pole position.

An asterisk in the "Geological Strike and dip" column indicates that the samples were cored perpendicular to the bedding plane. No entry in the same column indicates that the beds are flat-lying.

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TABLE 2.—LONGITUDES OF POLE POSITIONS DERIVED FROM EUROPEAN AND AMERICAN ROCKS OF THE SAME AGE

The longitudes for European rocks are the mean values for each age determined by various workers.

	European rocks Longitude	American rocks		Difference in longitude
		Formation	Longitude	
Triassic	147° E.	Chugwater	114° E.	+33°
		Moenkopi	110° E.	+37°
Permian	169° E.	Supai	104° E.	+65°
		Supai (Runcorn, 1956a)	119° E.	+50°
		Supai (Doell, 1955)	115° E.	+54°
Carboniferous	150° E.	Naco ss. (Runcorn, 1956a)	117° E.	+33°
		Naco ss. (C_P)	130° E.	+20°
		Deadwood (C_M)	117° E.	+33°
Cambrian	173° E.	Wilberns (Howell and Martinez, 1957)	158° E.	+15°
		Lodore	174° W.	−13°
Pre-Cambrian	123° W.	Hakatai shales (Doell, 1955)	127° W.	+4°
		Hakatai shales (Runcorn, 1956a)	150° W.	+27°
		Hakatai shales	174° W.	+51°
		McNamara	138° W.	+15°
		Miller Peak	166° W.	+43°
		Spokane (Prickly Pear Canyon)	152° W.	+29°
		Spokane (McDonald Creek)	139° W.	+16°
		Appekunny	156° W.	+33°
		Grinnell	152° W.	+29°
		Shinumo quartzite	173° W.	+50°
		Bonito Canyon quartzite	146° W.	+23°
		Bass limestone	158° W.	+35°

TABLE 3.—CALCULATED RATE OF POLAR WANDERING

Geological periods	Time interval in millions of years	Angular distance in degrees	Rate of movement in degrees per million years
Upper Triassic–Tertiary	150	28	0.2
Lower Triassic–Upper Triassic	30	10	0.3
Permian–Lower Triassic	30	12	0.4
Pennsylvanian–Permian	25	17	0.7
Silurian–Pennsylvanian	100	21	0.2
Ordovician–Silurian	25	14	0.6
Cambrian–Ordovician	90	20	0.2

TABLE 4.—SEQUENCE OF ALTERNATIONS OF MAGNETIZATION IN CHUGWATER AND MOENKOPI FORMATIONS

Chugwater Formation

A (Red Mtn.)		B (Rawlins)		C (Thermopolis)		D (Lander)		E (Dinwoody Lakes)		F (Fort Washakie)		G (Shell)		H (Sheep Mtn.)		I (Alcova Resvr.)		J (Troublesome Creek)	
Samples	Sign	Samples	Sign	Samples	Sign	Samples	Sign	Samples	Sign	Samples	Sign	Samples	Sign	Samples	Sign	Samples	Sign	Samples	Sign
										10–11	−ve								
12–18	+ve			6	+ve	7–10	+ve	9–10	+ve	9	+ve								
10–11	−ve	10	−ve	3–5	−ve	5–6	−ve	7–8	−ve	6–8	−ve								
3–9	+ve	3–9	+ve	1–2	+ve	1–4	+ve	3–6	+ve	1–5	+ve	2–6	+ve	1–10	+ve	1–10	+ve	19–27	+ve
1	−ve	1–2	−ve					1–2	−ve			1	−ve						

Moenkopi Formation

A (Capitol Reef)		B (Sand Canyon)		C (Split Mtn.)		D (Vernal)		E (Poverty Tank)		F (Echo Cliffs)	
Samples	Sign	Samples	Sign	Samples	Sign	Samples	Sign	Samples	Sign	Samples	Sign
										24	+ve
		8	−ve					13–15	−ve	11–23	−ve
2–10	+ve	1–7	+ve	1–6	+ve	1–9	+ve	12	+ve	8–10	+ve
								1–11	−ve	3–7	−ve

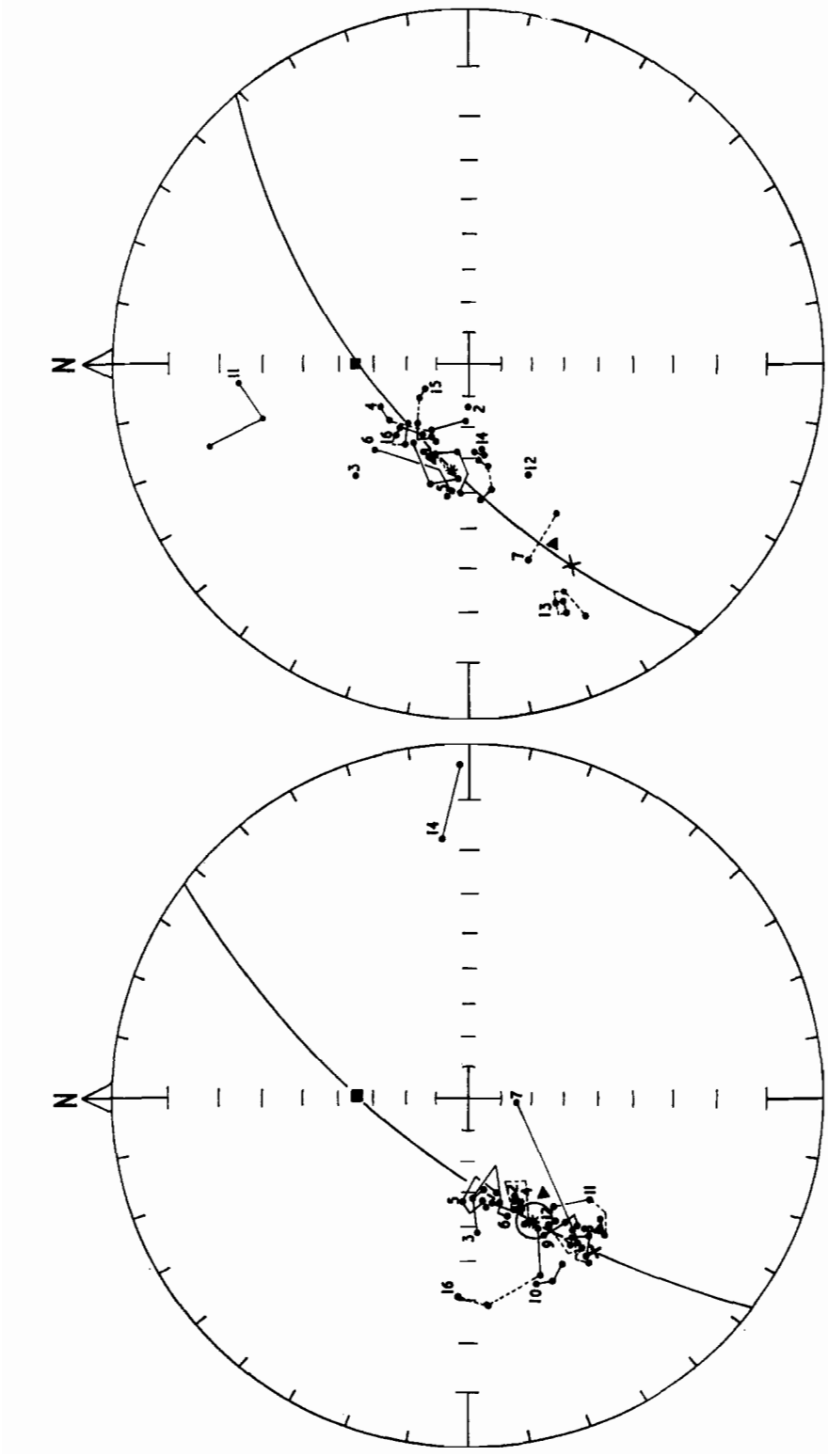


FIGURE 1
FIGURE 1.—BASS LIMESTONE (ALGONKIAN) FROM KAIBAB TRAIL, S. RIM OF GRAND CANYON
FIGURE 2. HAKATAI SHALES (ALGONKIAN) FROM KAIBAB TRAIL, N. RIM OF GRAND CANYON

FIGURE 3

FIGURE 4

FIGURE 3.—SHINUMO QUARTZITE (ALGONKIAN) FROM KAIBAB TRAIL, S. RIM OF GRAND CANYON
FIGURE 4.—BONITO CANYON QUARTZITE (PRECAMBRIAN) FROM BLUE CREEK, FORT DEFIANCE, ARIZONA

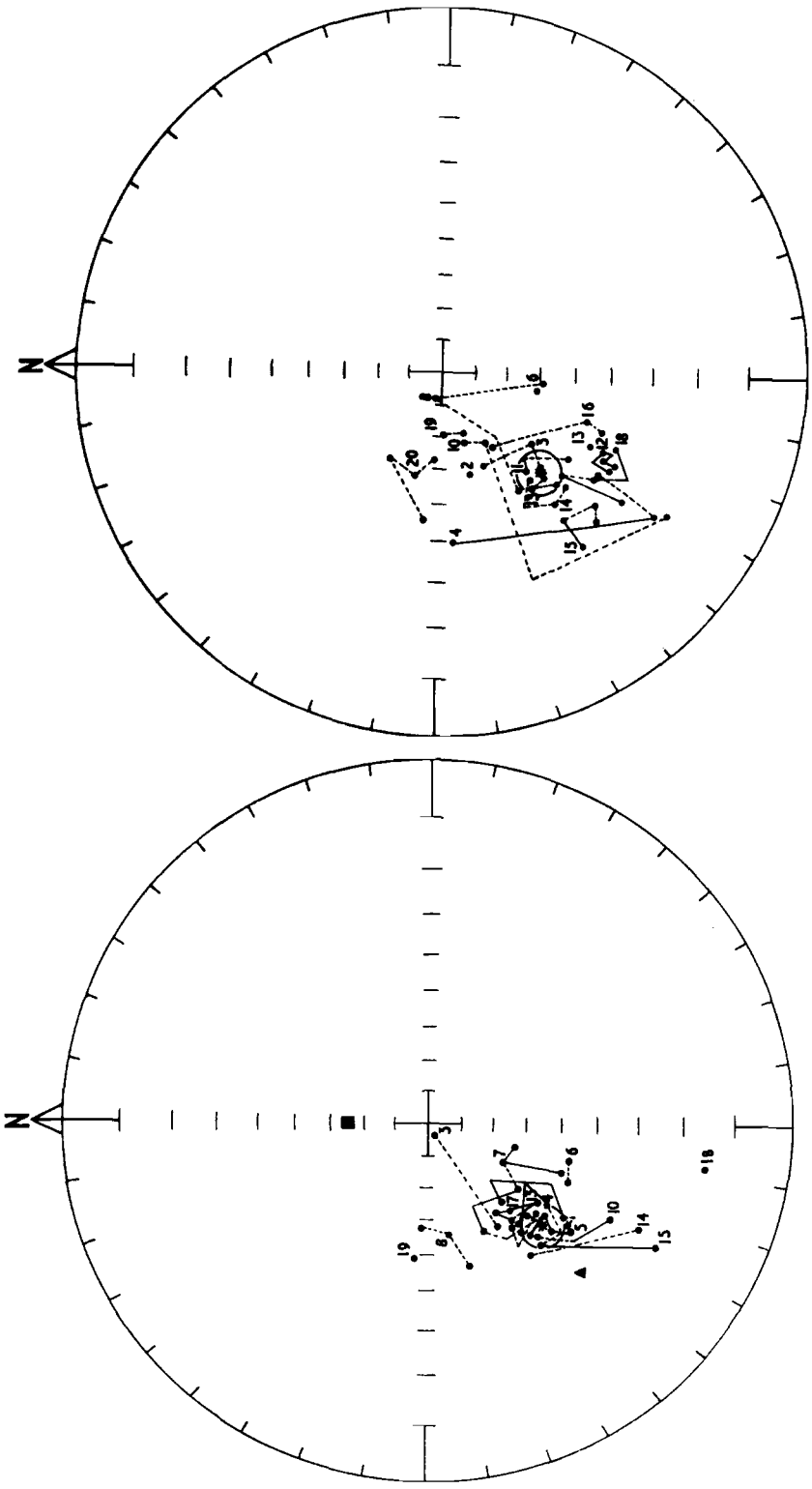


FIGURE 5
FIGURE 6
FIGURE 5.—BELT SERIES (ALGONKIAN): APPEKUNNY FORMATION FROM GLACIER NATIONAL PARK, MONTANA
FIGURE 6.—BELT SERIES (ALGONKIAN): GRINNELL FORMATION FROM GLACIER NATIONAL PARK, MONTANA

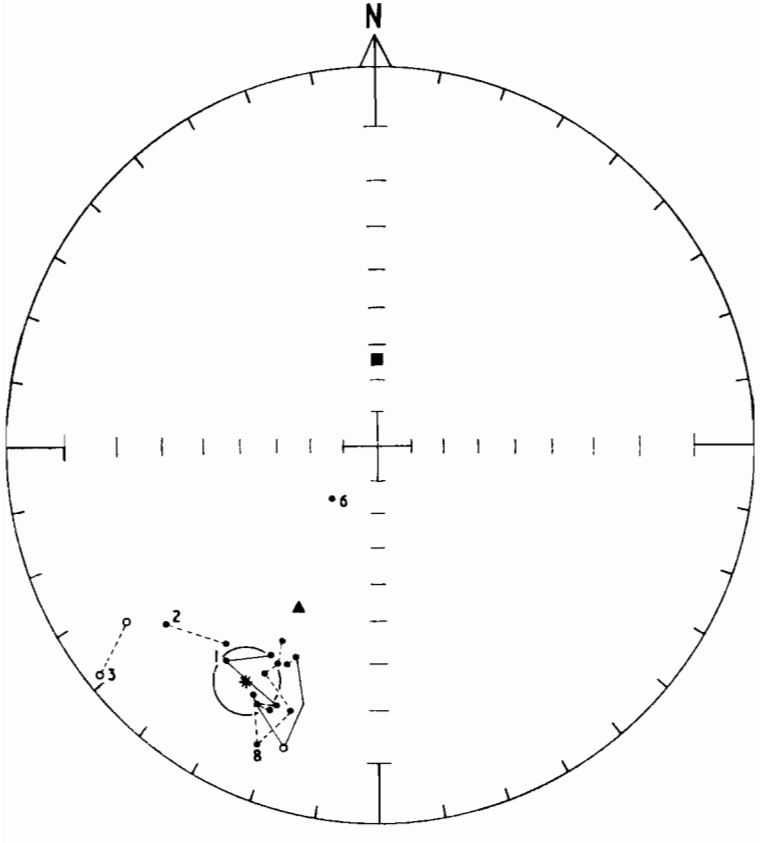


FIGURE 7

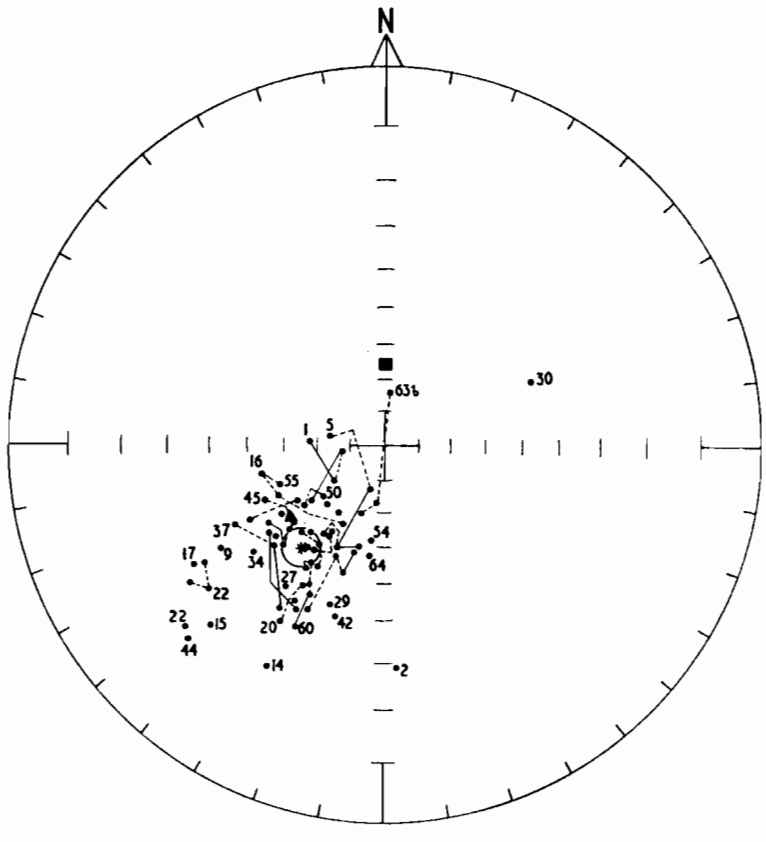


FIGURE 8

FIGURE 7.—BELT SERIES (ALGONKIAN): SPOKANE FORMATION FROM MCDONALD CREEK, MONTANA
FIGURE 8.—BELT SERIES (ALGONKIAN): SPOKANE FORMATION FROM PRICKLY PEAR CANYON, MONTANA

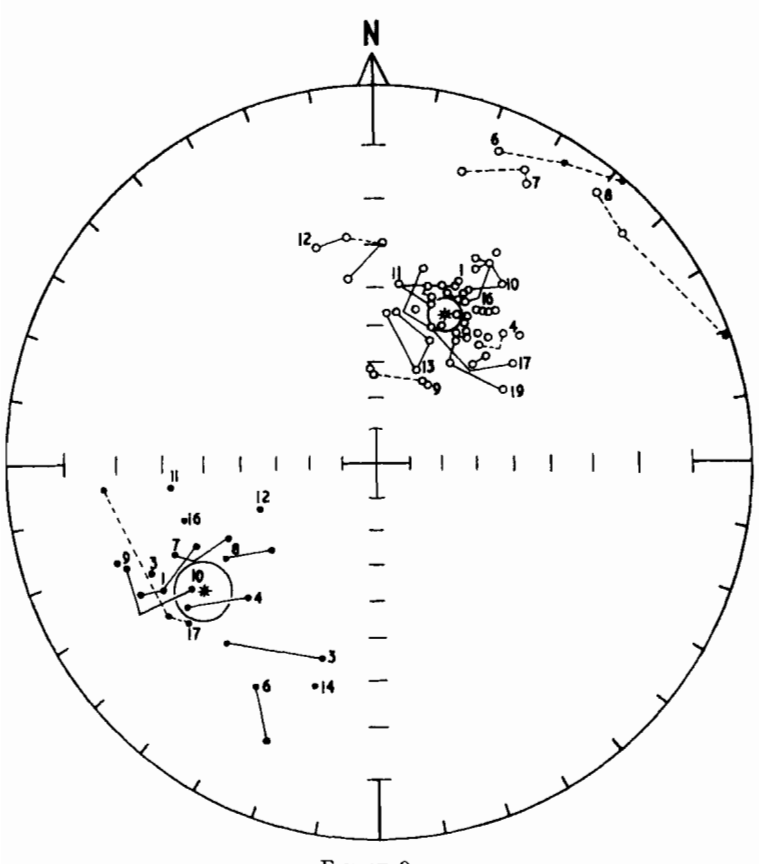


FIGURE 9

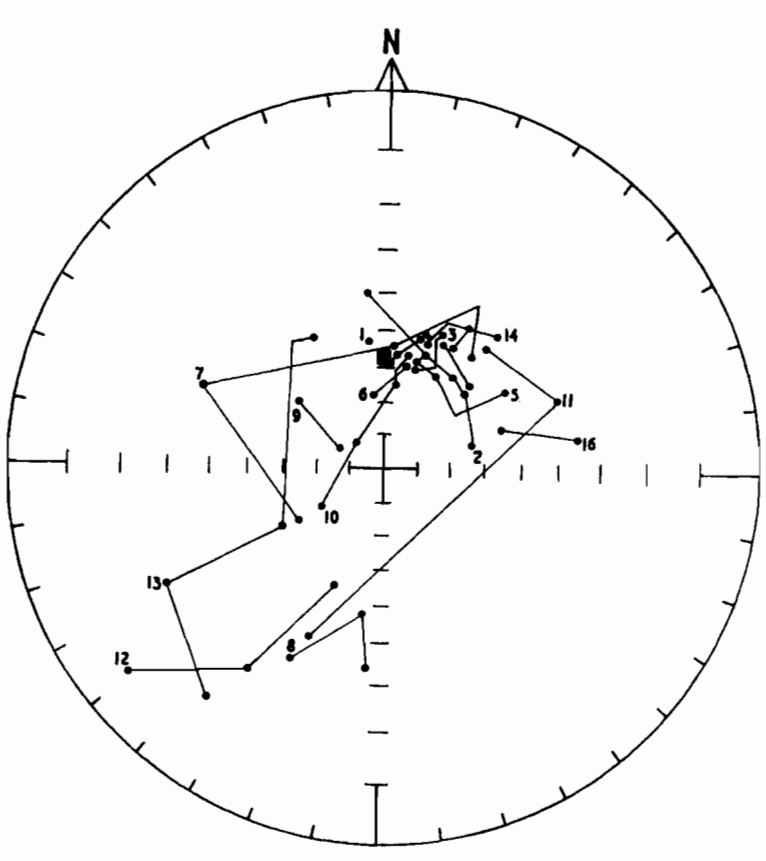


FIGURE 10

FIGURE 9.—BELT SERIES (ALGONKIAN)

N. E. −ve magnetizations: McNamara formation from Blackfoot Canyon, Missoula, Montana. S. W. +ve magnetizations: Miller Peak formation from Donovan Creek, east of Missoula, Montana.

FIGURE 10.—TAPEATS SANDSTONE (CAMBRIAN) FROM PEACH SPRINGS, ARIZONA

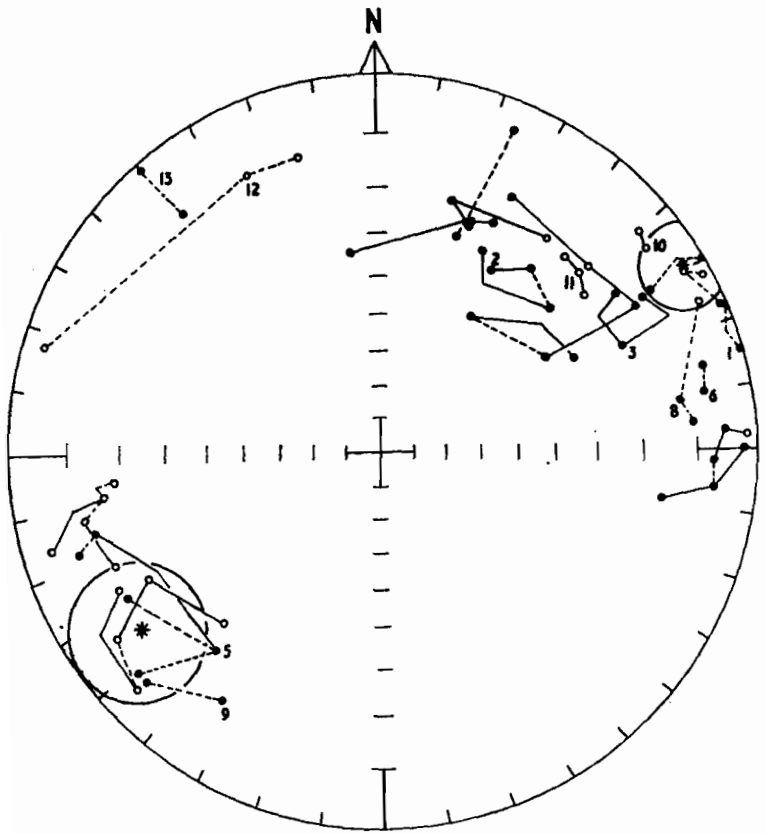


FIGURE 11

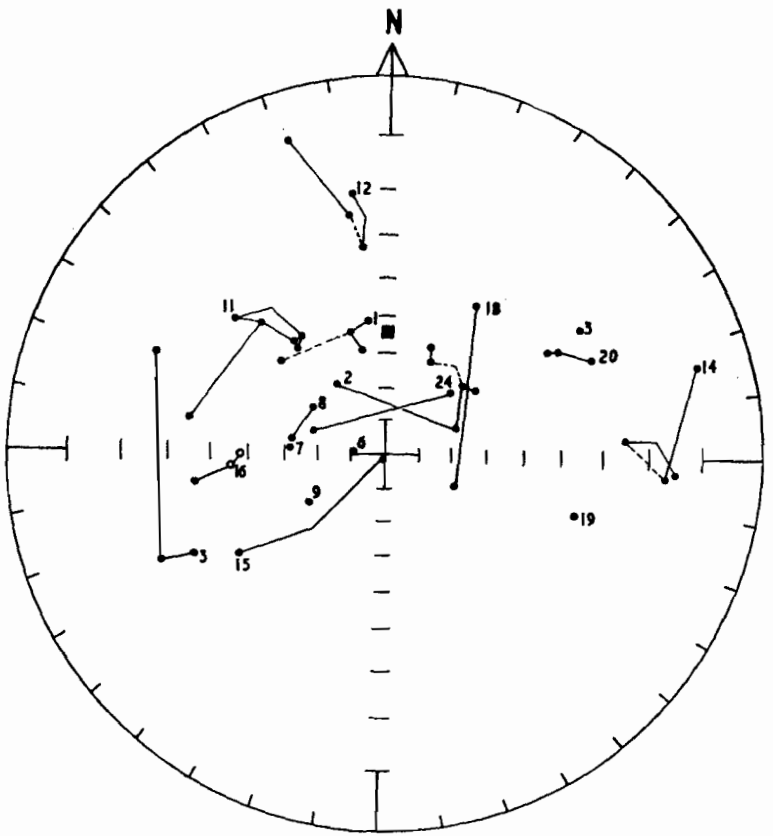


FIGURE 12

FIGURE 11.—BRIGHT ANGEL SHALE (CAMBRIAN) FROM DIAMOND BAR RANCH, ARIZONA
FIGURE 12.—LODORE FORMATION (CAMBRIAN) FROM UINTA MOUNTAINS, UTAH

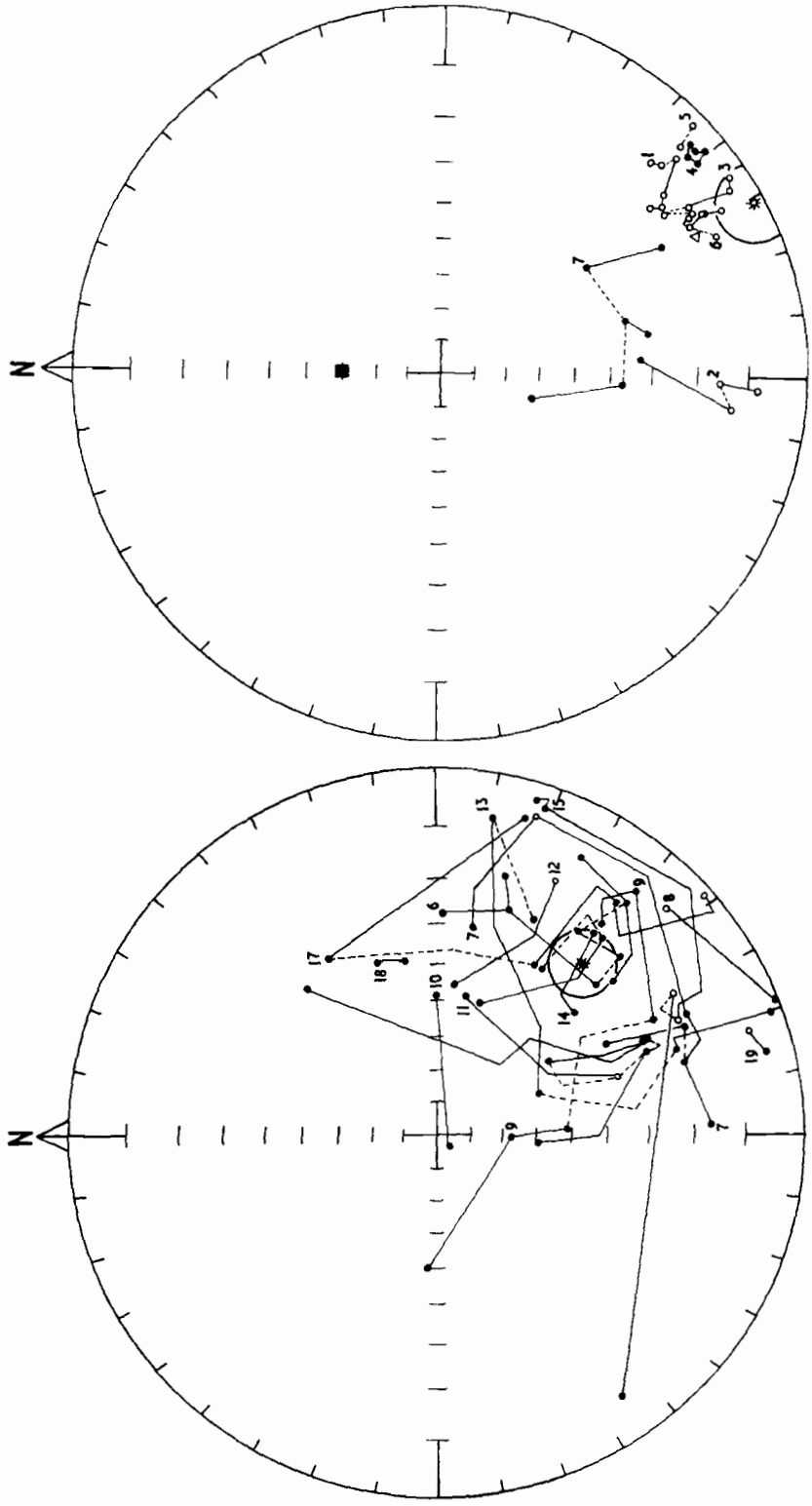


FIGURE 13

FIGURE 14

FIGURE 13.—JUNIATA BEDS (ORDOVICIAN) FROM BEDFORD, PENNSYLVANIA
FIGURE 14.—DEADWOOD FORMATION (MISSISSIPPIAN) FROM RAWLINS, WYOMING

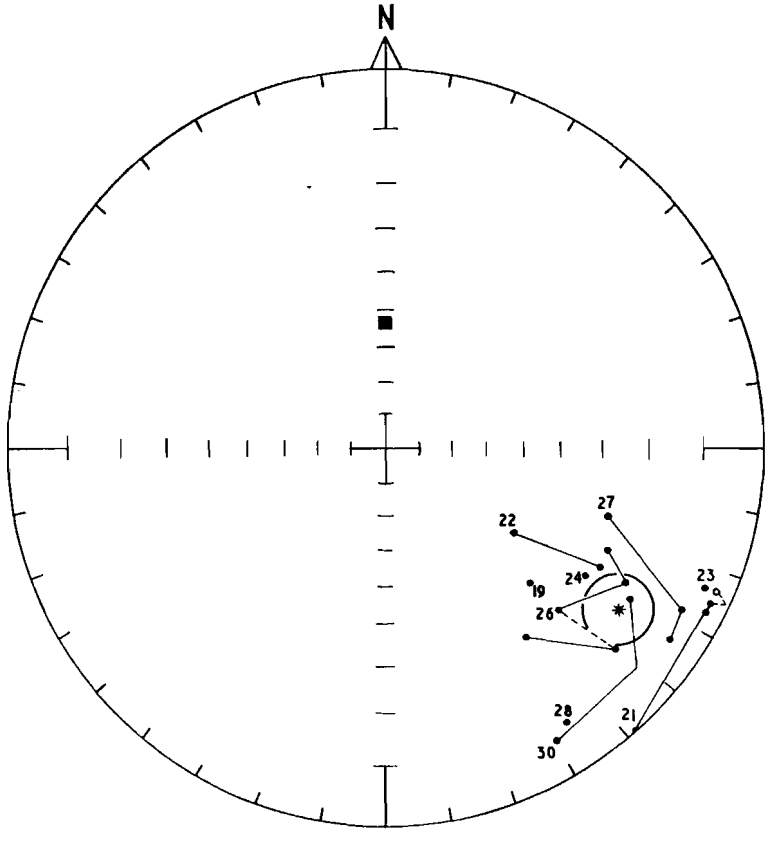
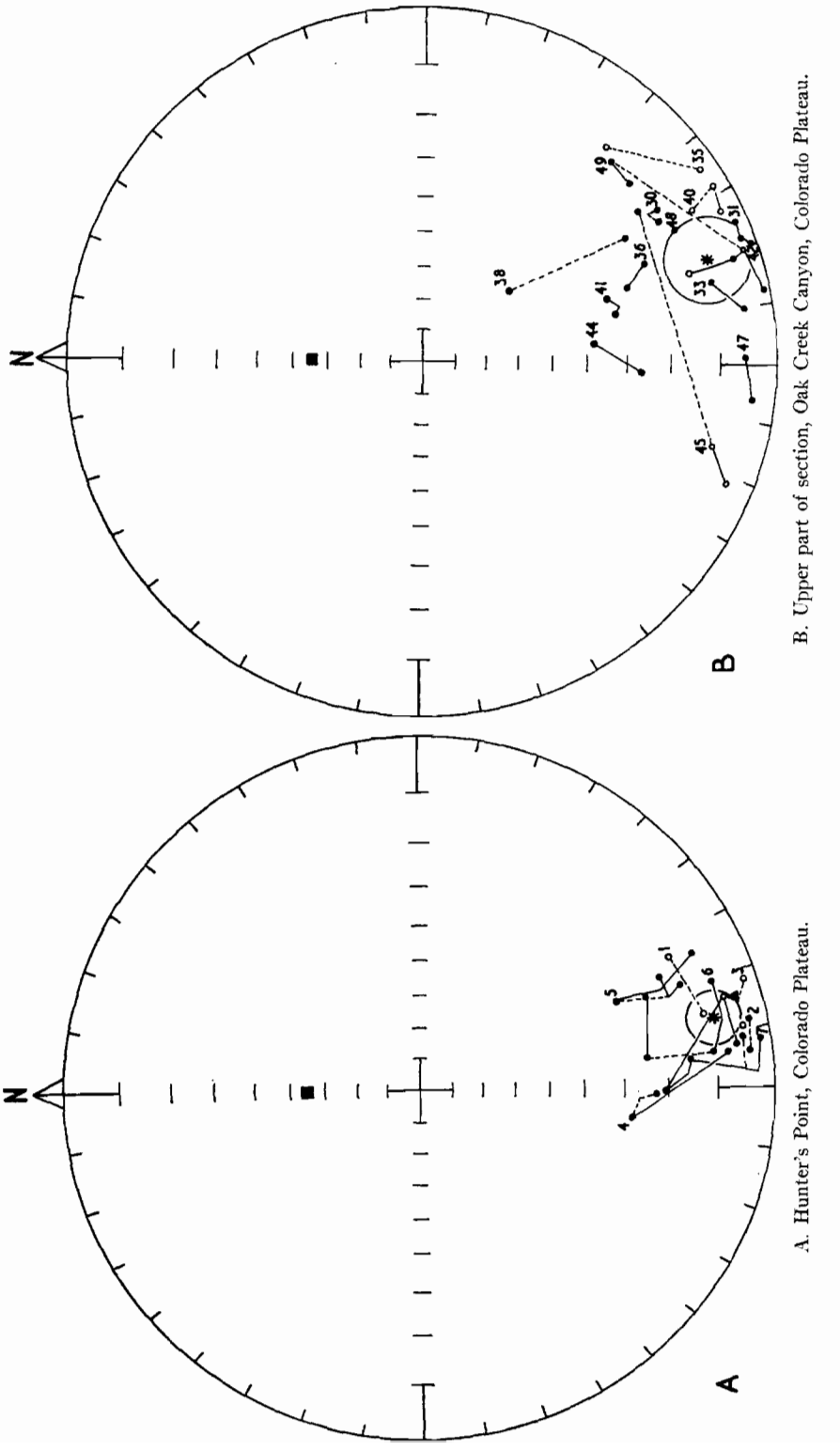
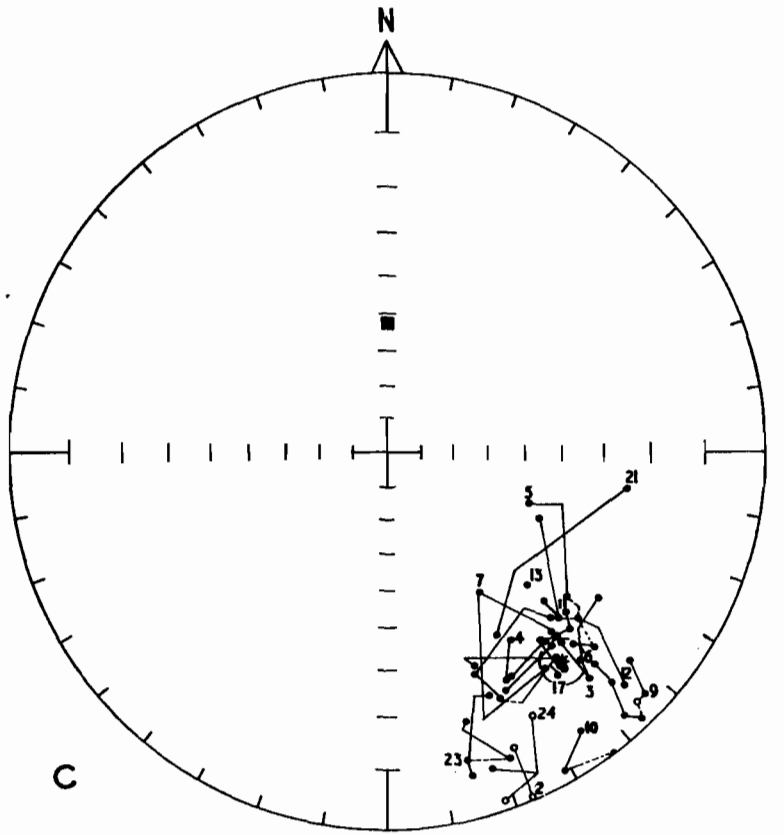


FIGURE 15.—NACO SANDSTONE (PENNSYLVANIAN) FROM COLORADO PLATEAU



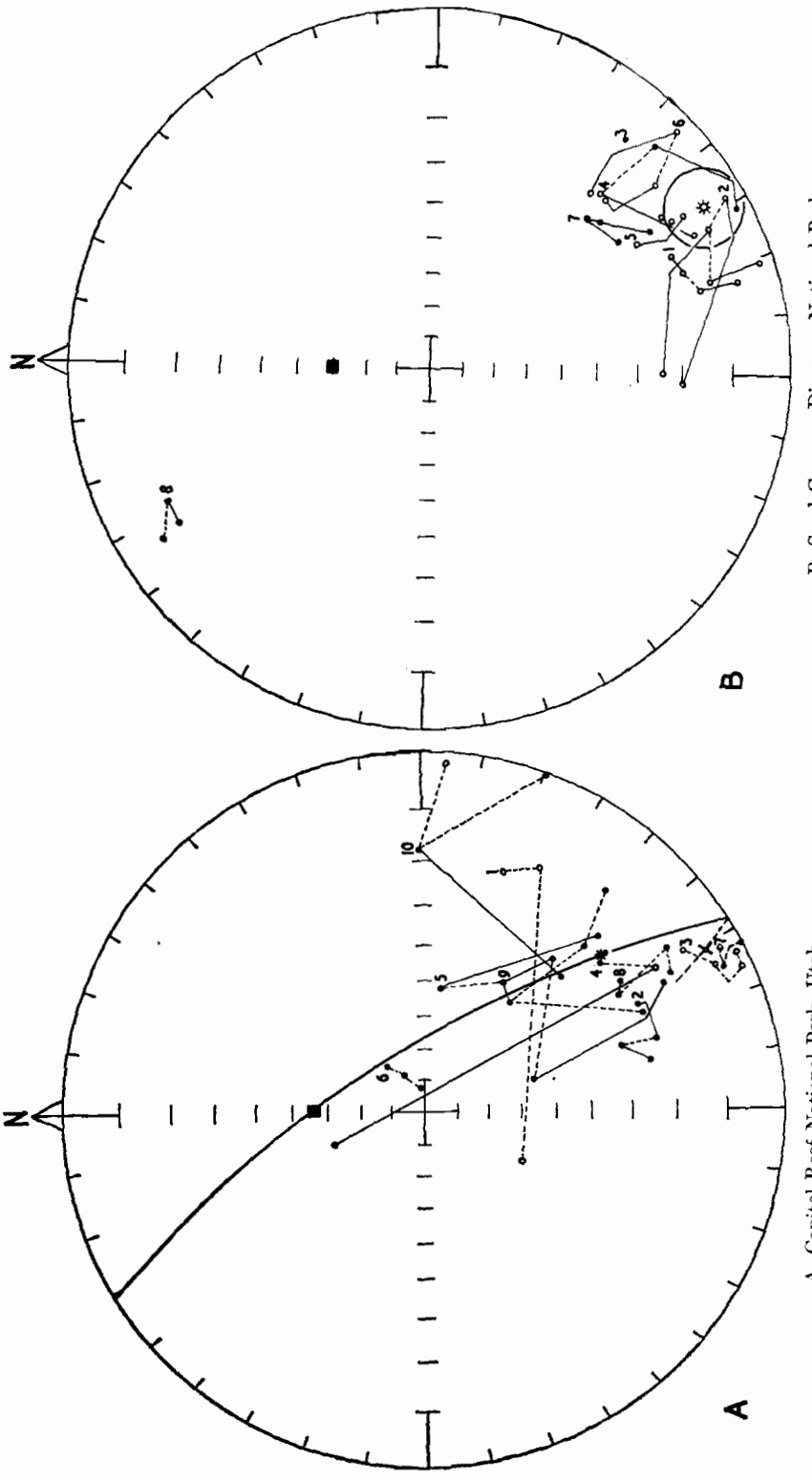
A. Hunter's Point, Colorado Plateau.

B. Upper part of section, Oak Creek Canyon, Colorado Plateau.



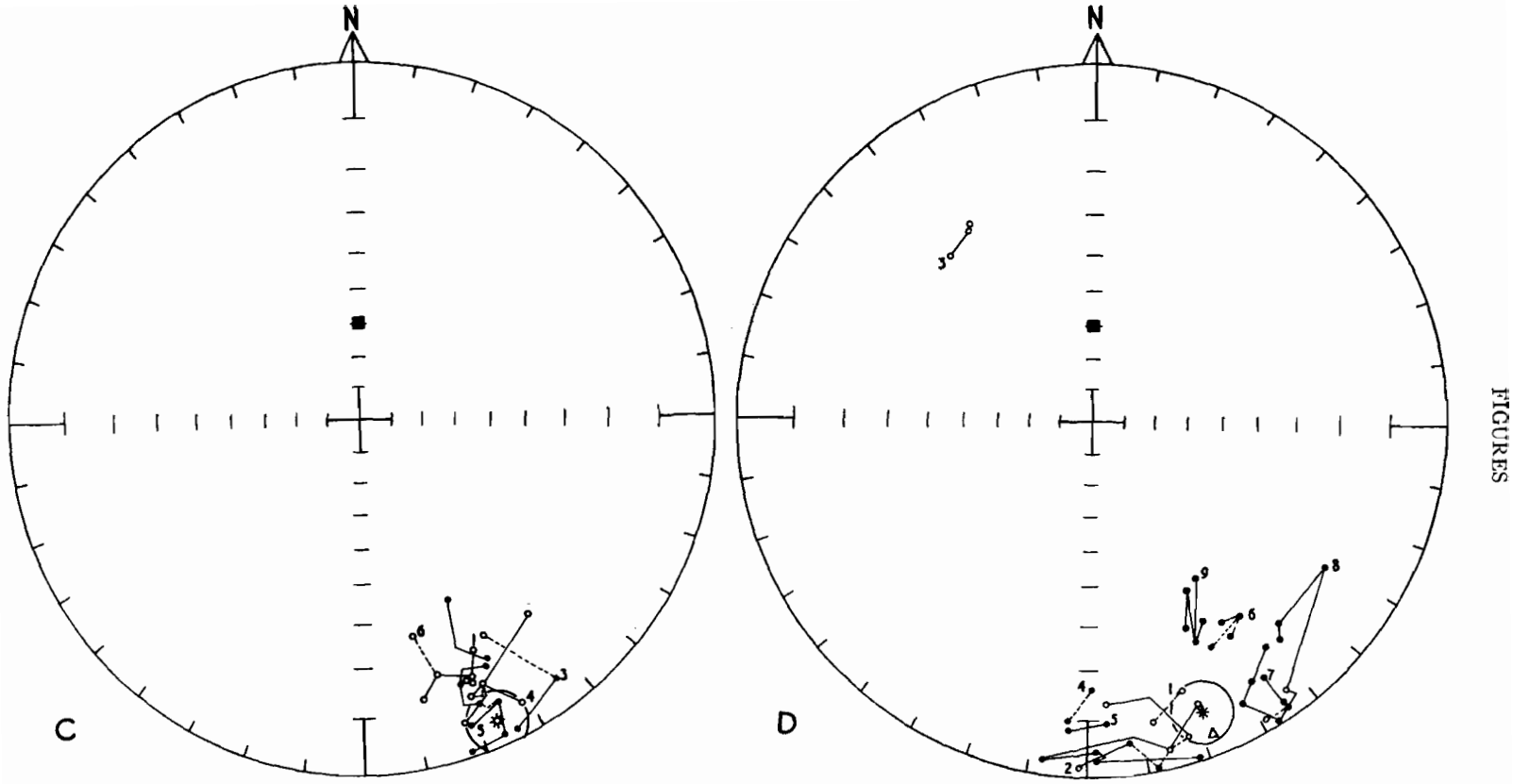
C. Lower part of section, Oak Creek Canyon, Colorado Plateau.

FIGURE. 16—SUPAI FORMATION (PERMIAN)



A. Capitol Reef National Park, Utah.

B. Sand Canyon, Dinosaur National Park.

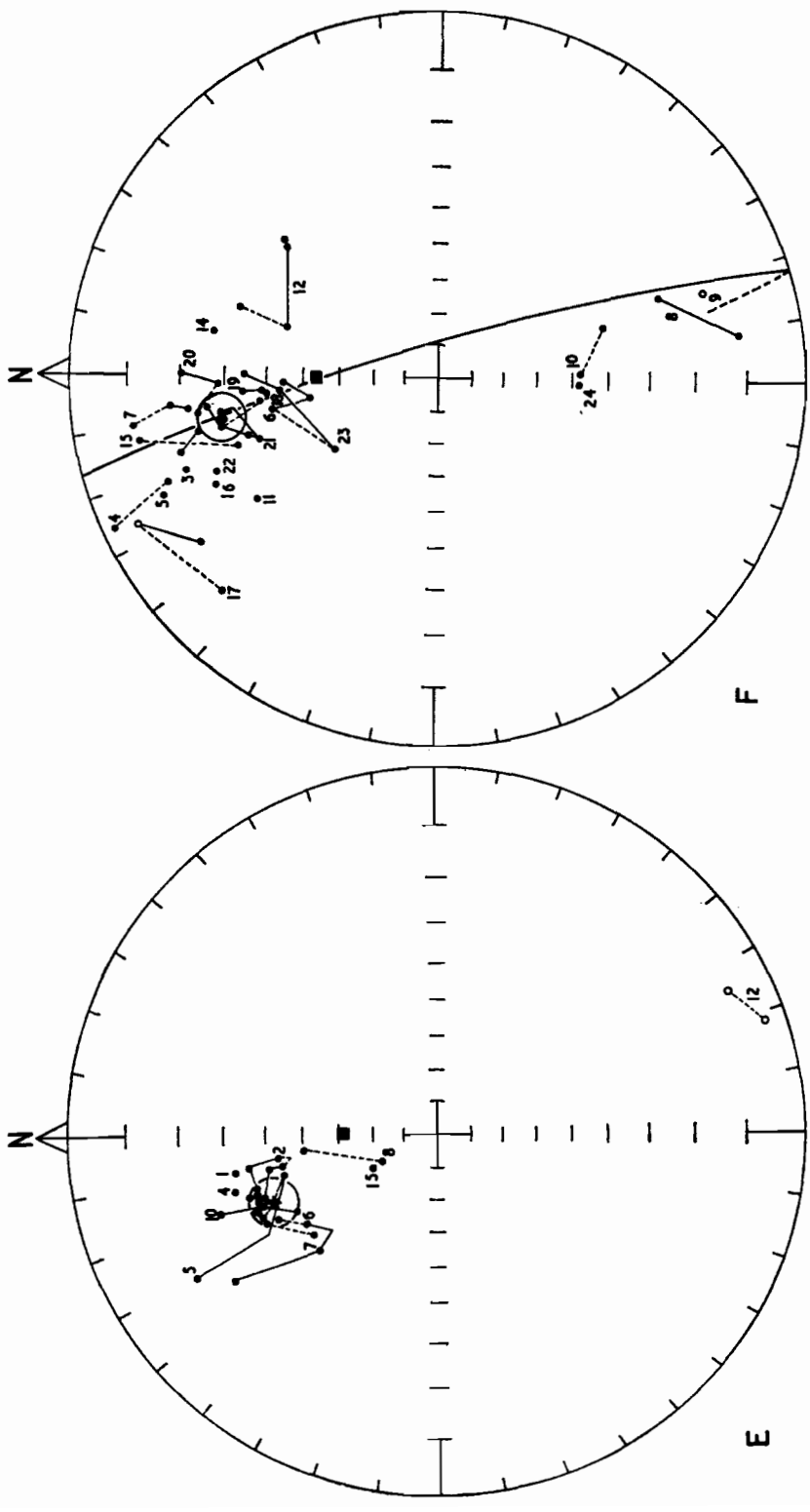


C. Split Mountain, Colorado.

D. Vernal, Utah.

FIGURE 17.—MOENKOPI FORMATION (LOWER TRIASSIC)

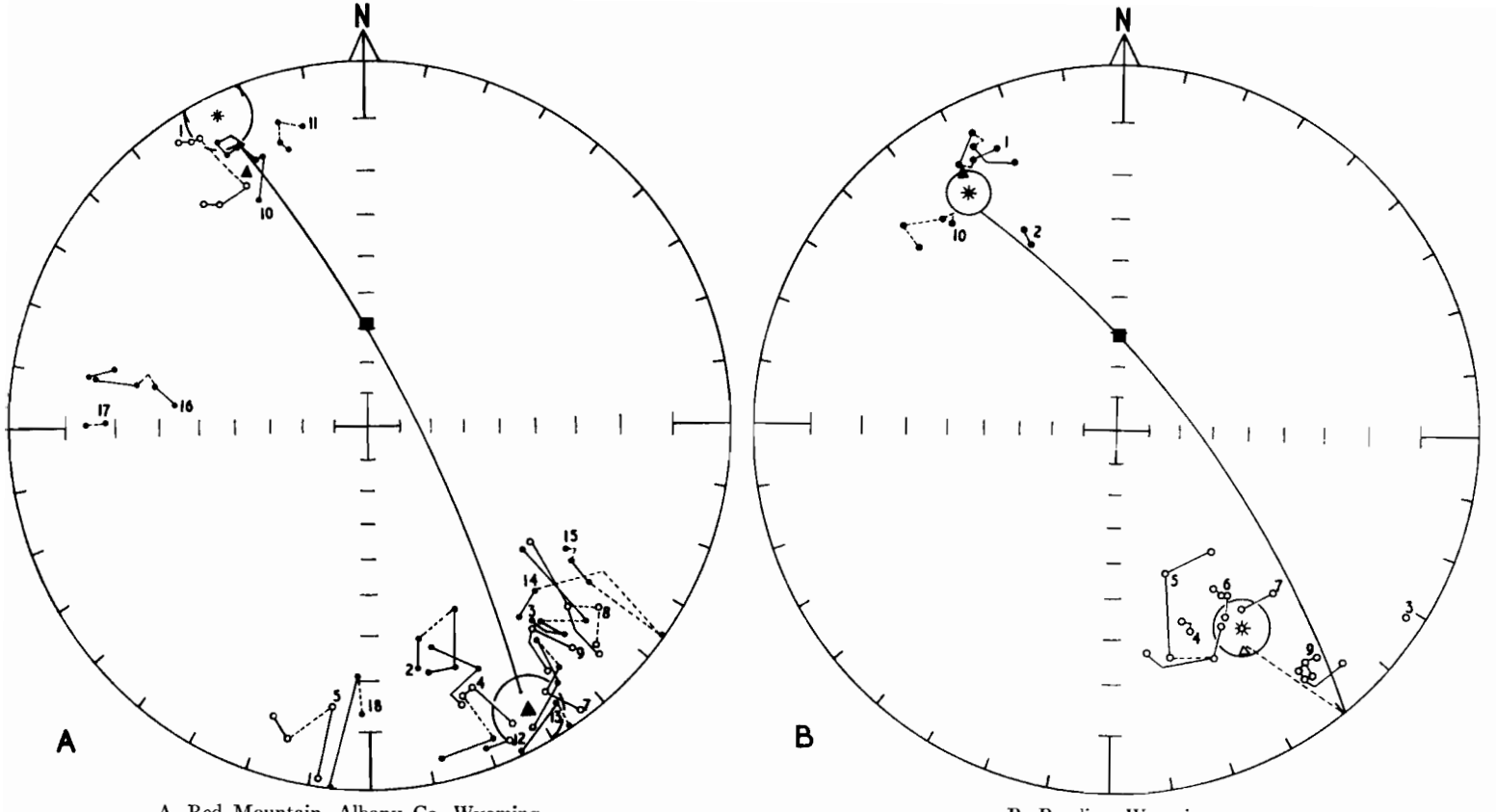
(*Continued on next page*)



E. Poverty Tank, Colorado Plateau.

F. Echo Cliffs, Colorado Plateau.

FIGURE 17. (*Cont'd.*)

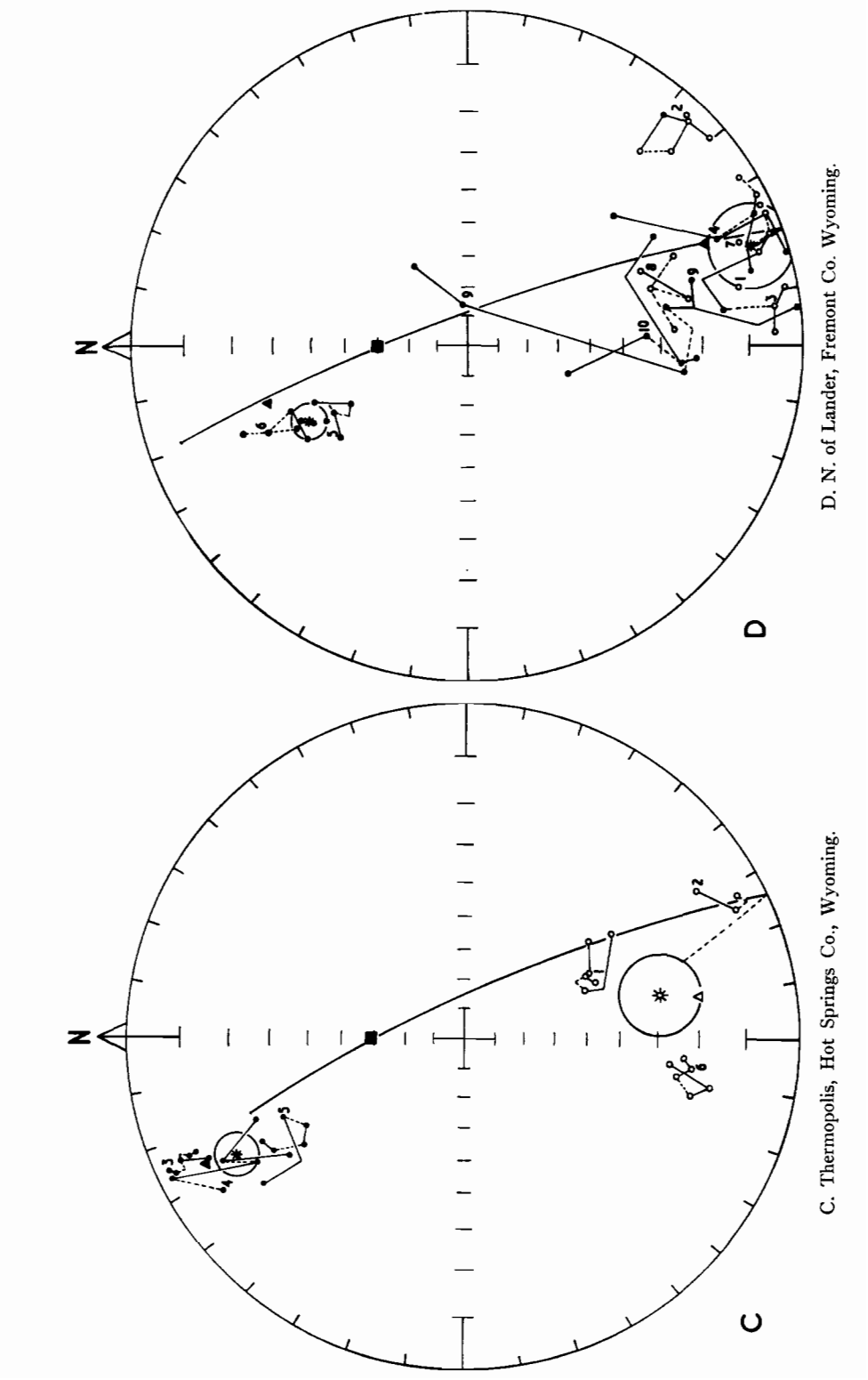


A. Red Mountain, Albany Co. Wyoming.

B. Rawlins, Wyoming.

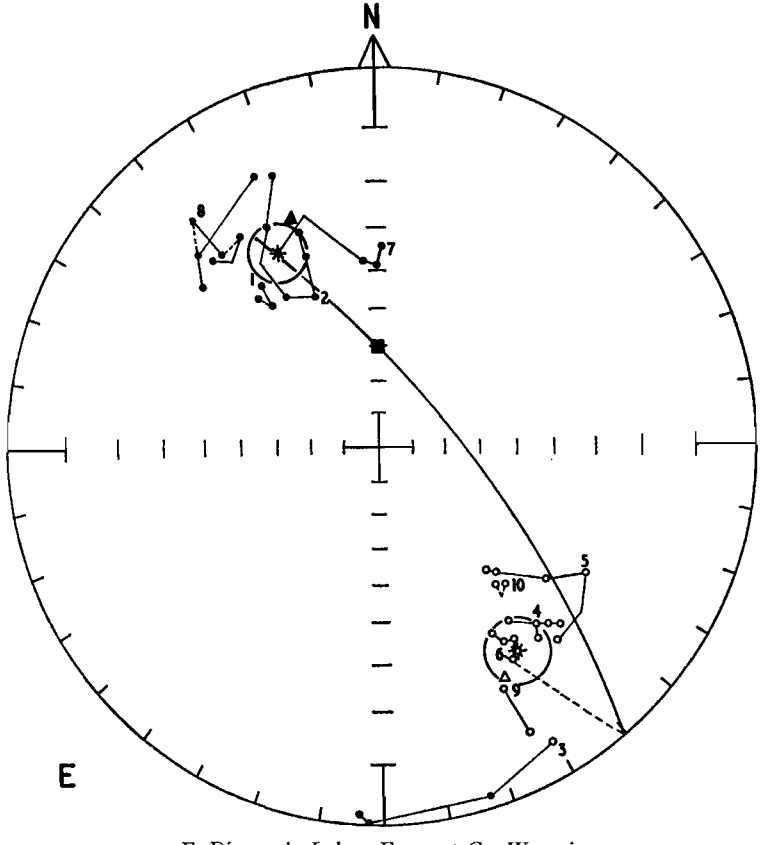
FIGURE 18.—CHUGWATER FORMATION (TRIASSIC)

(Continued on following pages)

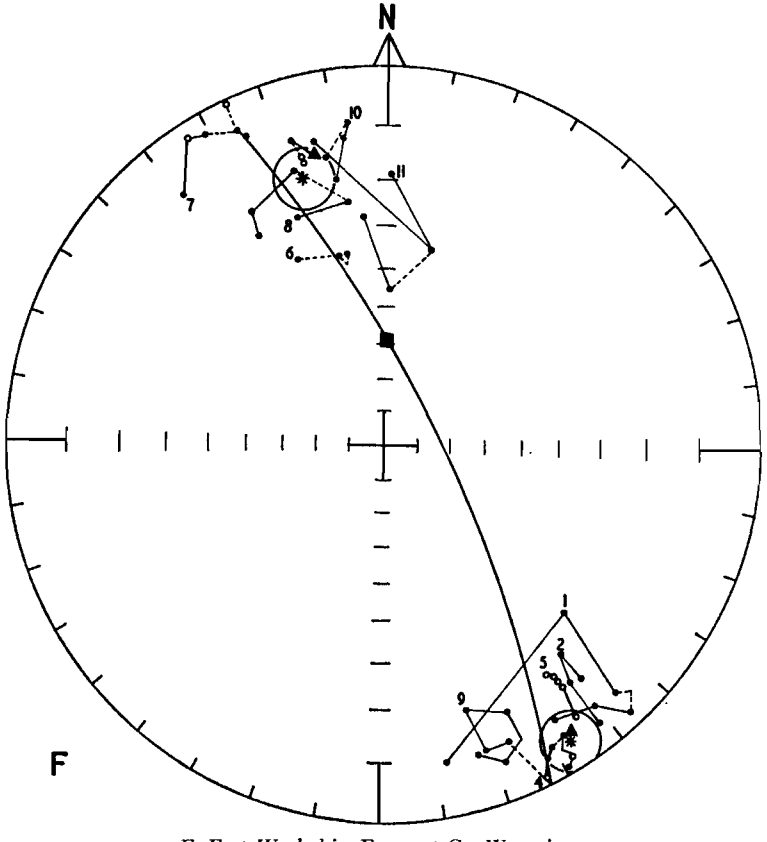


D. N. of Lander, Fremont Co. Wyoming.

C. Thermopolis, Hot Springs Co., Wyoming.

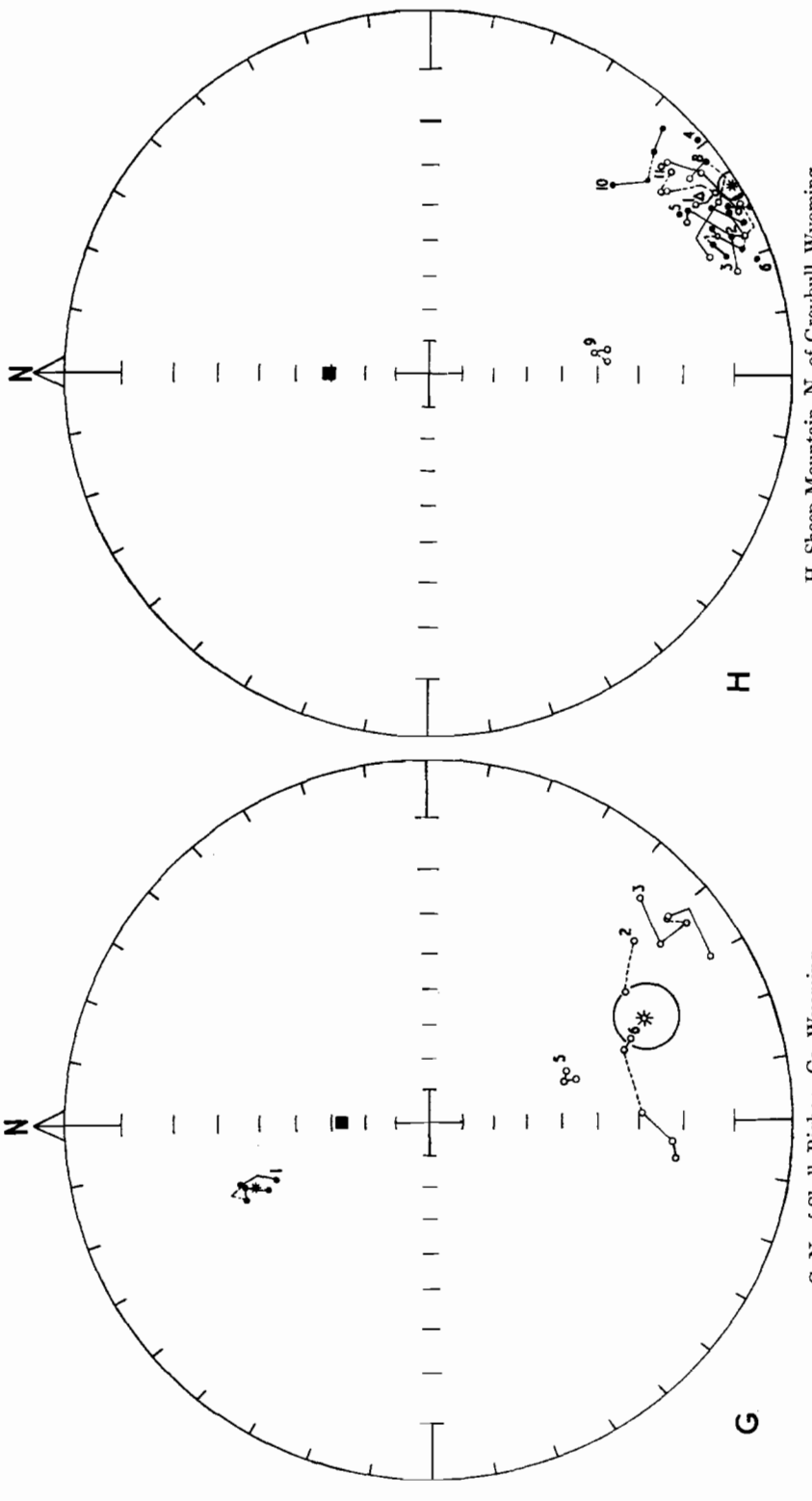


E. Dinwoody Lakes, Fremont Co. Wyoming.



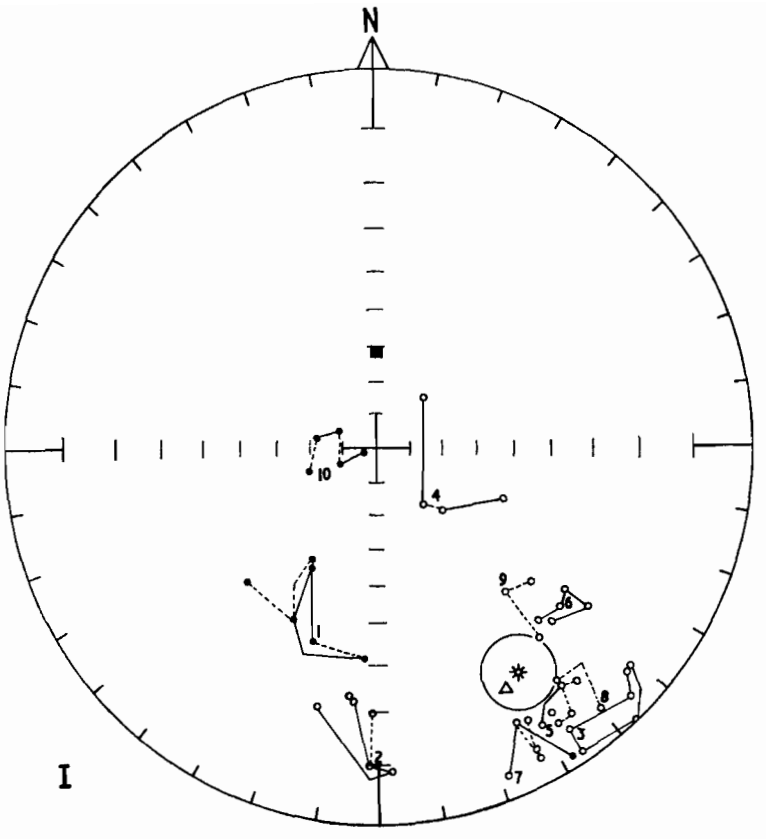
F. Fort Washakie, Fremont Co. Wyoming.

FIGURE 18 (*Cont'd.*)

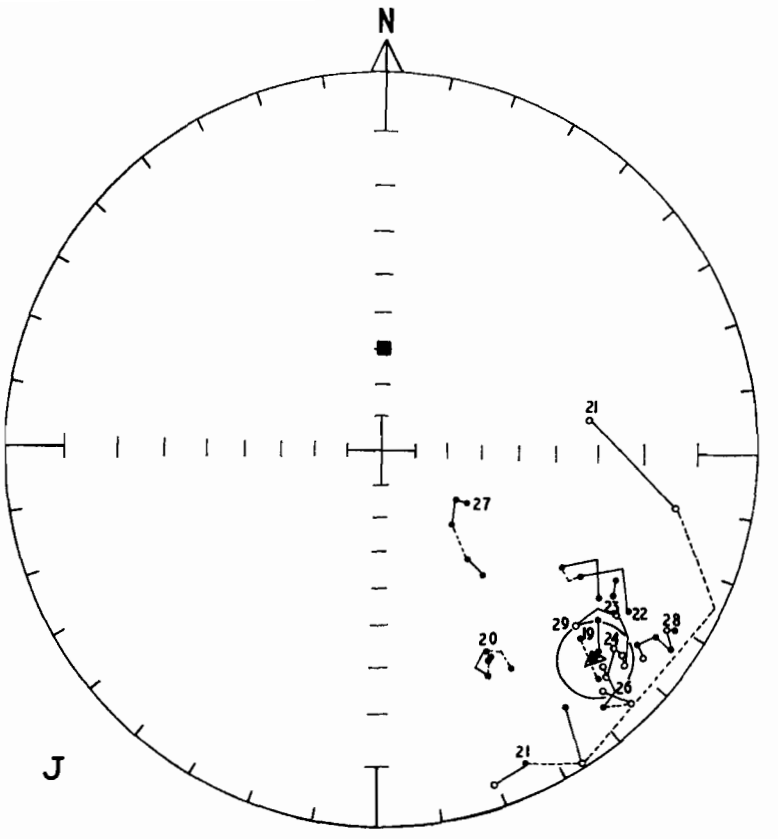


H. Sheep Mountain, N. of Greybull, Wyoming.

G. N. of Shell, Bighorn Co. Wyoming.



I. Alcova Reservoir, N. of Rawlins, Wyoming.



J. Troublesome Creek, Carbon Co. Wyoming.

FIGURE 18. (*Cont'd.*)

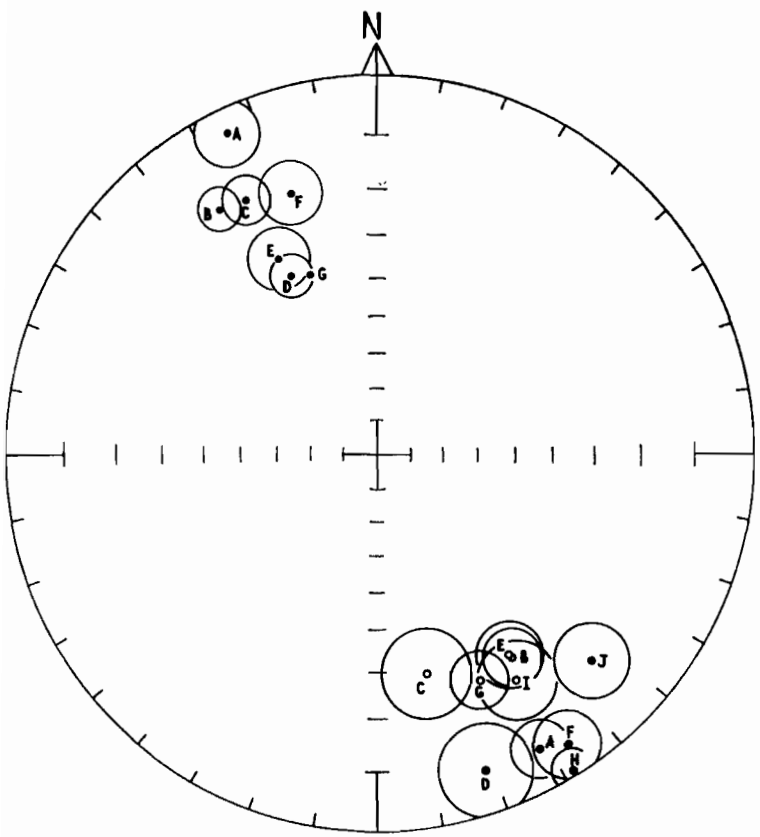


FIGURE 19

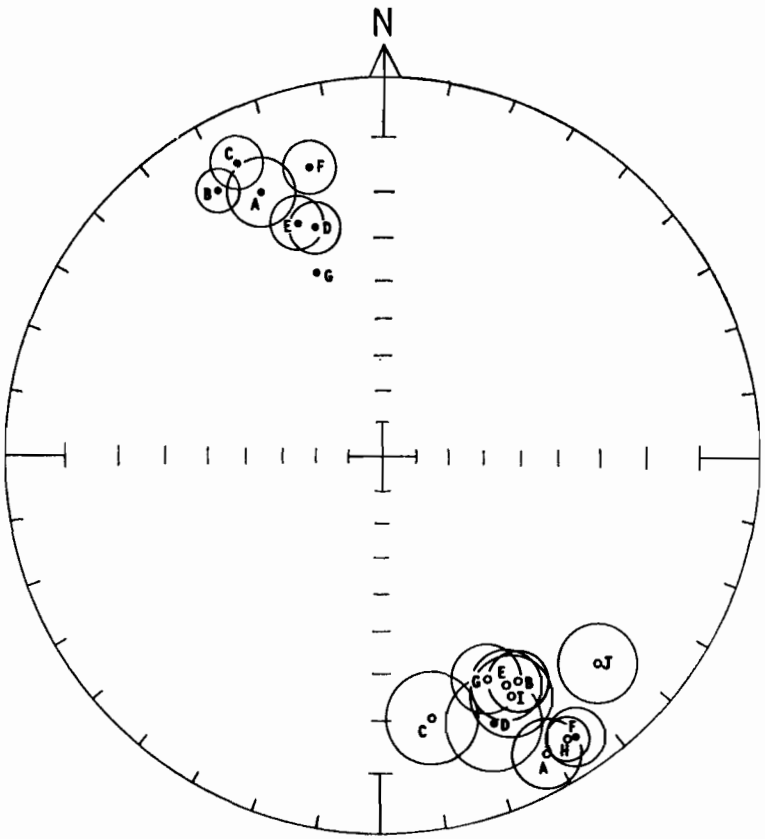


FIGURE 20

FIGURE 19.—MEAN DIRECTIONS OF MAGNETIZATION OF CHUGWATER FORMATION BEFORE CORRECTION FOR GEOLOGICAL DIP OF THE BEDS
Symbols as in Figure 18

FIGURE 20.—MEAN DIRECTIONS OF MAGNETIZATION OF CHUGWATER FORMATION AFTER CORRECTION FOR GEOLOGICAL DIP OF THE BEDS
Symbols as in Figure 18

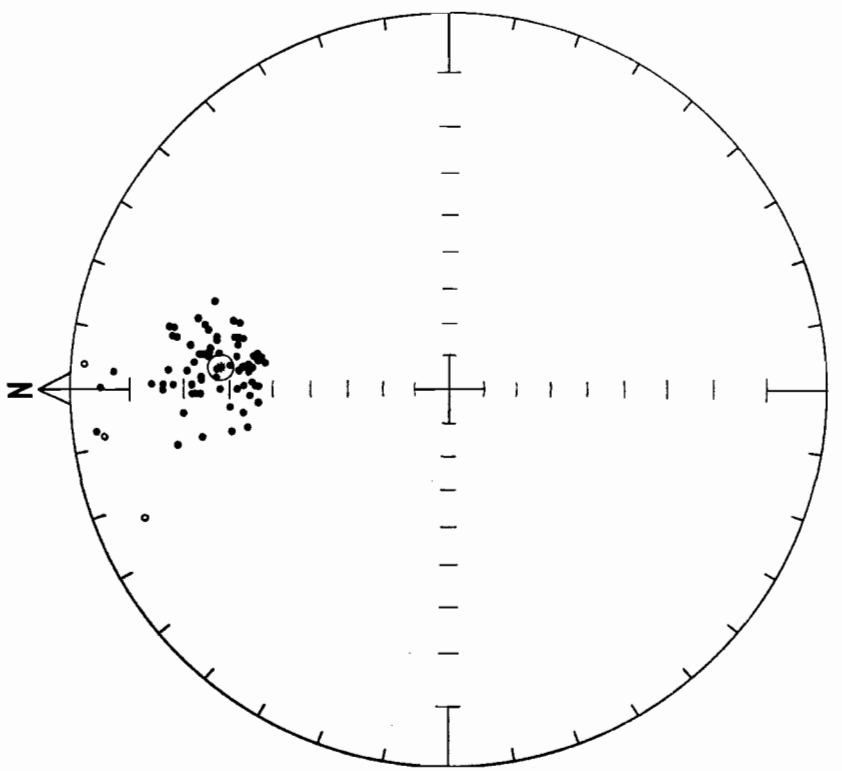
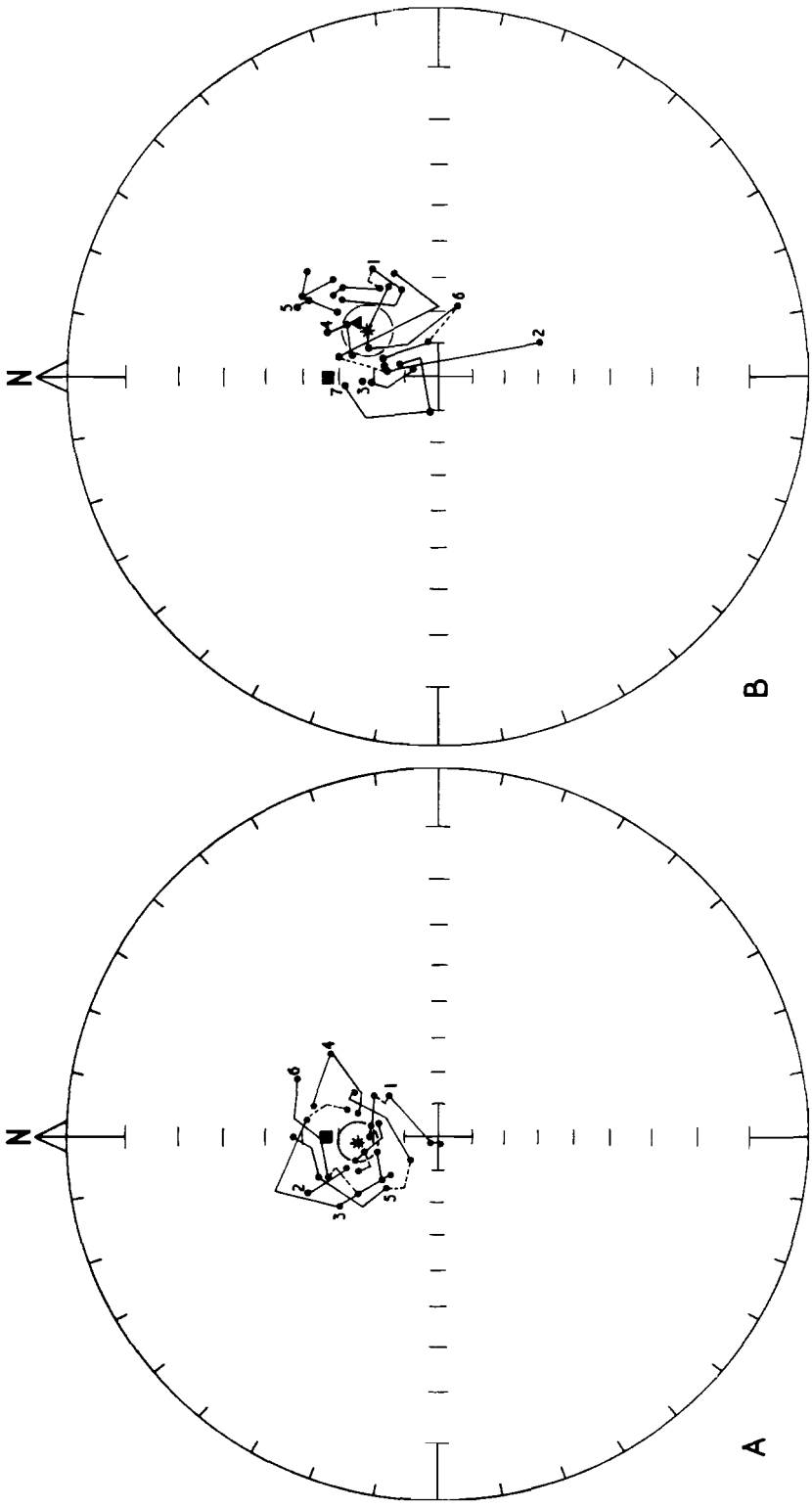
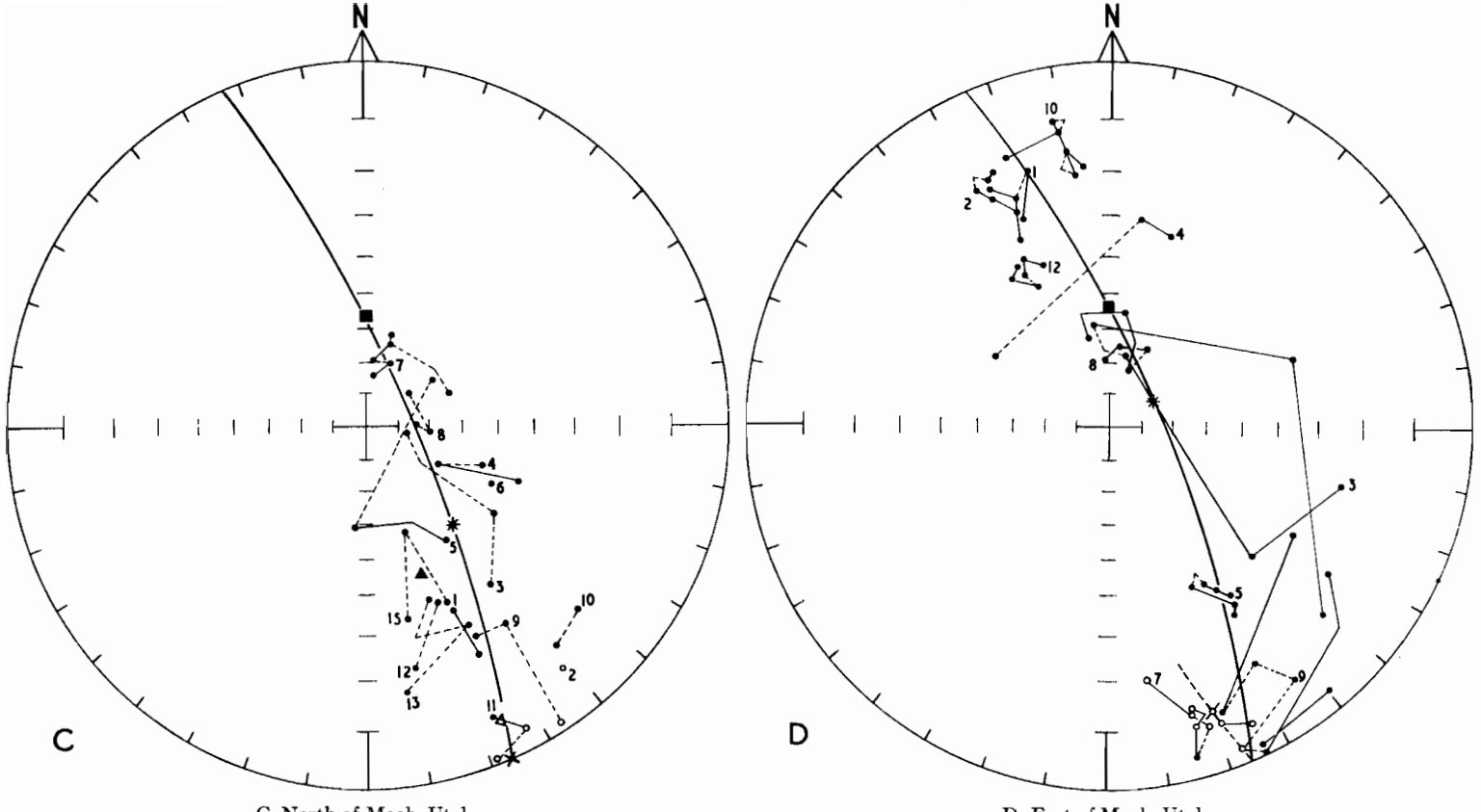


FIGURE 21.—NEWARK GROUP (TRIASSIC) FROM FRENCHTOWN, NEW JERSEY



A. Site I at Colorado National Monument

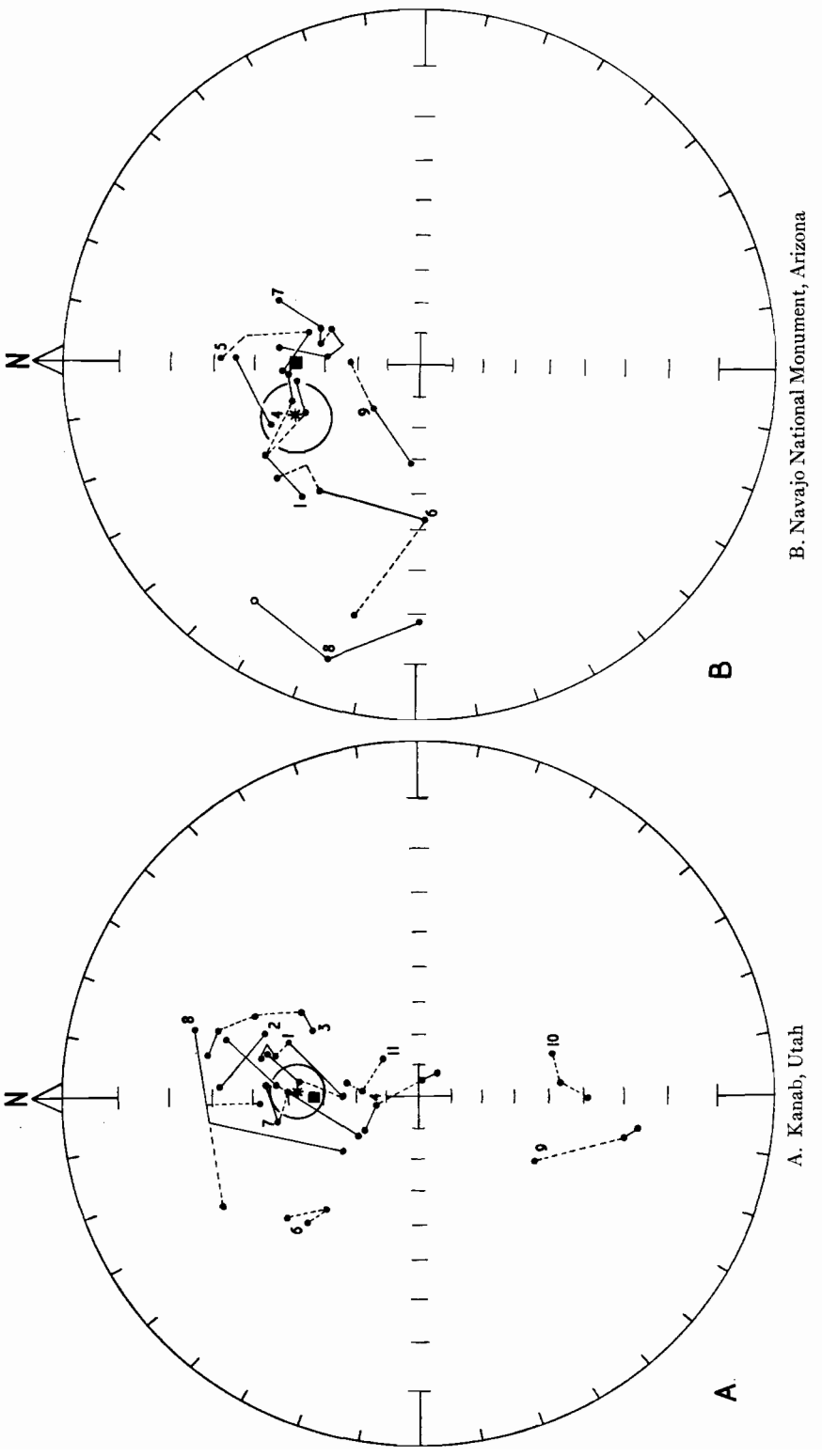
B. Site II at Colorado National Monument

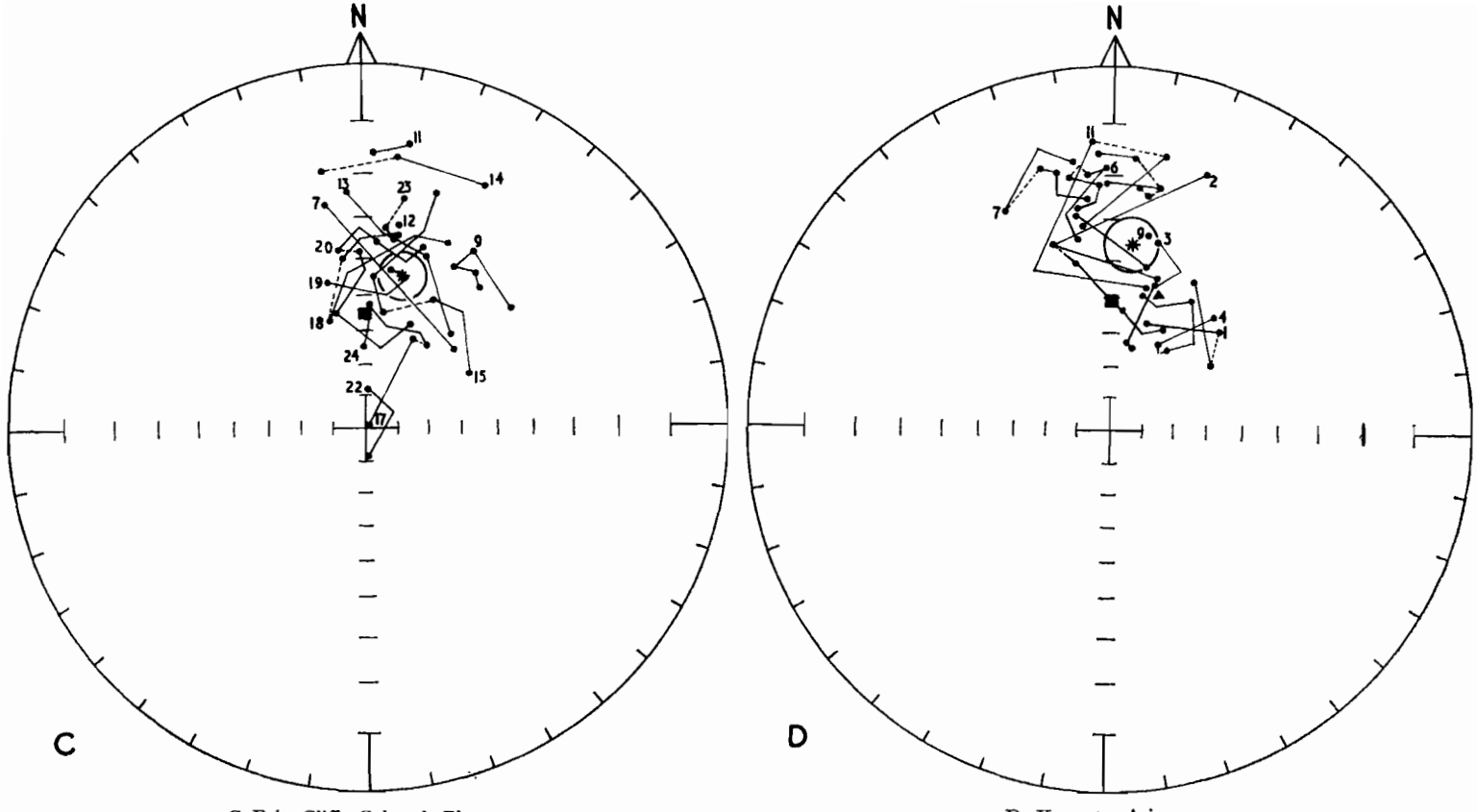


C. North of Moab, Utah

D. East of Moab, Utah

FIGURE 22.—CHINLE FORMATION (UPPER TRIASSIC)


N
A
A. Kanab, Utah
N
B
B. Navajo National Monument, Arizona



C. Echo Cliffs, Colorado Plateau

D. Kayenta, Arizona

FIGURE 23.—KAYENTA FORMATION (LOWER JURASSIC)

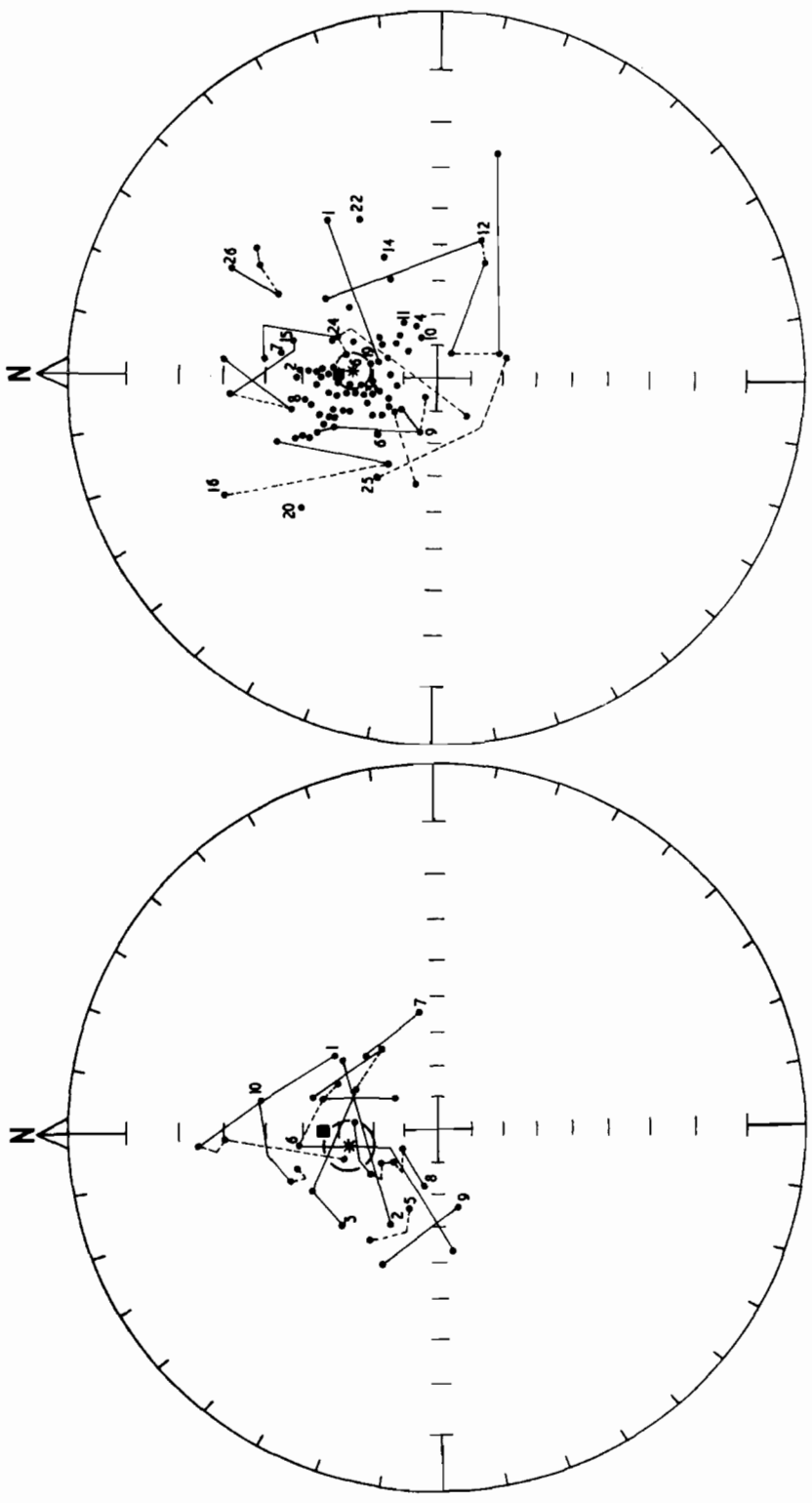
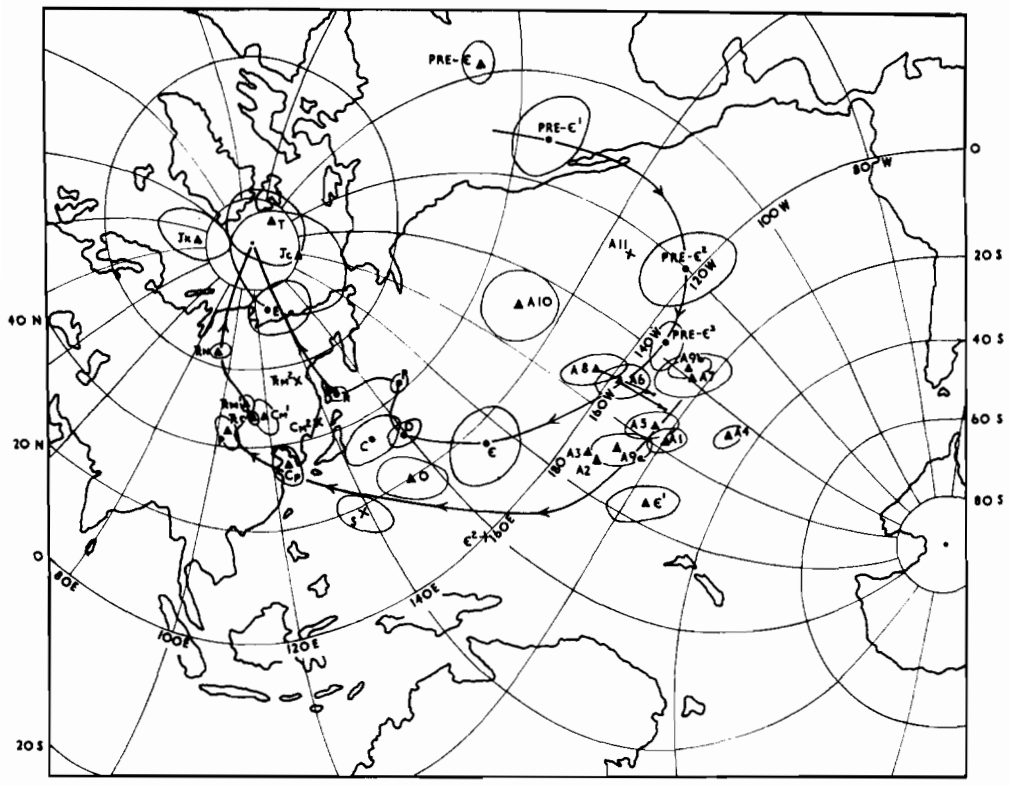


FIGURE 24
FIGURE 24.—CARMEL FORMATION (JURASSIC) FROM MOAB, UTAH
FIGURE 25.—DUCHESNE RIVER FORMATION (TERTIARY) FROM ROOSEVELT, UTAH

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PRE-€
PRE-€¹
PRE-€²
PRE-€³
120W
140W
160W
180
160E
140E
120E
100E
80E
100W
80W
0
20 S
40 S
60 S
80 S
40 N
20 N
0
20 S
A10
A11
A8
A6
A9b
A7
A5
A4
A1
A3
A2
A9a
A0
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Downloaded from gsabulletin.gsapubs.org on October 21, 2010

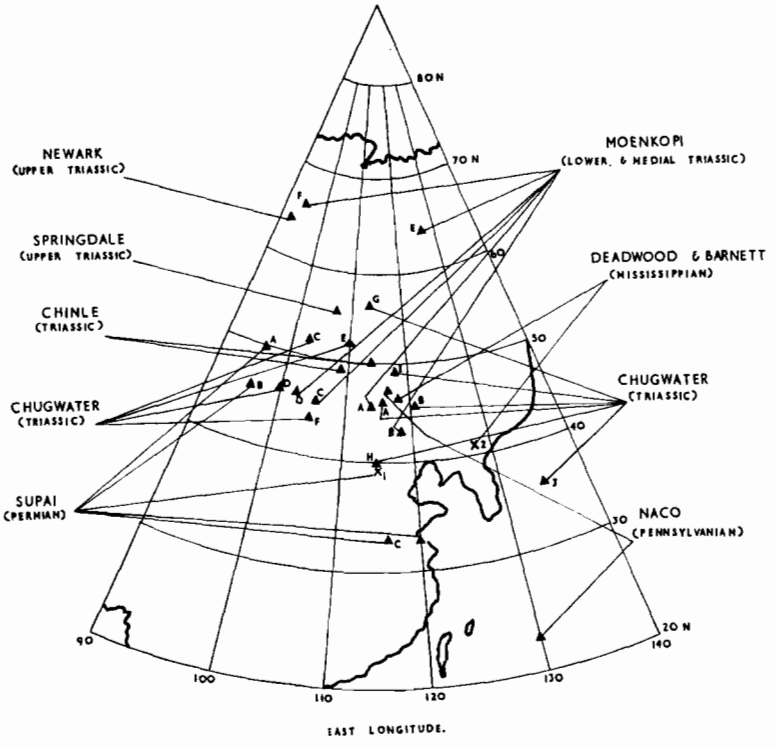


FIGURE 27.—POLE POSITIONS FOR TRIASSIC, PERMIAN, AND CARBONIFEROUS ROCKS

Letters on the Moenkopi, Chugwater, and Supai pole positions refer to localities listed in the text of this paper. Other pole positions denoted by a triangle represent pole positions determined in this paper and in Runcorn (1956a). Pole positions determined by a cross refer to other authors as follows: (1) Doell (1955); (2) Howell and Martinez (1957). (Stereographic polar projection)